Who Writes the Answer Key?

Lessons from Forty-Eight Rounds of Trying to Benchmark Autonomous Physics Research

Erin Krasniqi

Independent researcher

Working paper, version 1, October 2026

Drafted by Claude (Anthropic) under the author's direction and reviewed by ChatGPT Pro (OpenAI); roles and methods are in Appendix F.

Abstract

We report a month-long attempt (23 September to 21 October 2026) to build an automatically scored benchmark of autonomous scientific research in physics that models could not saturate without genuinely improving. The project was conceived and led by one human, who set the goal, approved every model run, carried every message between three AI systems (ChatGPT Pro as designer, Codex as builder and runner, and Claude as director and verifier), and supplied several of the ideas the designs were built on. Over about 48 rounds the team built and tested many designs:

- toy discovery tasks and constructive research with a checker;
- generated worlds with disguised scientific revolutions, twin laws and hidden unifications;
- calibrated far forecasts;
- clue worlds;
- multi-layer discovery horizons;
- physics truths that need large computation;
- method invention scored exactly.

In 162 subject-model episodes, 123 of them with gpt-6.1-sol, that model solved nearly every single-principle discovery world we scored (23 of the 49 scored cells supplied the law or a hint, or were controls). Its one systematic failure, far forecasts without the candidate law family, was not shown to be specific to the model: four generic public-information baselines, tested after the fact, did worse overall. Every design was eventually rejected. Read in hindsight, many rejections fall into four overlapping groups by how the answer key was built, although some had more local causes (a reference that did not converge, a design assumption that failed, a clue printed on the task card, prior art). Keys built from the designers' knowledge were reached by the solver; a trace audit could not tell whether by recall or by fitting candidate families. Authored laws were found by the model, or their scored behavior was learned by generic programs. Computed truths could support only a small separation on one machine, and we derive a cap on that separation. In the verifier families, standard methods reached certified optima at small sizes; in the last family, repeated runs of one method agreed without an optimality certificate, and we closed it by decision. Producing answer keys was easy; we did not establish an in-house task family that combined validated scoring, attainability from the supplied evidence, resistance to the declared shortcuts, and useful headroom for the tested model. We state this as an empirical conjecture about design teams with resources like ours, with a test that would count against it, and we place the known constructions that look like exceptions. Outside sources of keys (researchers' unpublished results, nature, much larger compute) are worth exploring, but our record establishes neither that outside provenance is necessary nor that it is sufficient. We state three rules for benchmark builders: programs reject and models certify; a speed gap is not research headroom; measure the gap at the scored precision. We document the AI designers' own failures, several of which resemble capabilities the benchmark sought to measure, and sketch a protocol for outside answer keys together with its failure modes.

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1 Introduction

Most benchmarks for AI systems pose a problem whose answer is already known and check whether the system finds it. Scientific research works differently. A researcher decides what to measure or compute, notices when the accepted description no longer fits, proposes a better one, and stakes it on predictions that nobody has checked yet. A benchmark that measured this whole loop, and that a model could not pass without actually becoming better at research, would be valuable. This paper reports a month-long attempt to build one, why every design failed, and what the failures teach.

The goal came from one person, Erin, who led the project. Their first message, a draft brief shared with ChatGPT Pro on 23 September 2026, asked for a very hard benchmark built on complex work such as physics, one that would test intuition and whether a model can make the kind of "jump" that humans make. Nearly four weeks later they summed it up in one line: a benchmark of full, autonomous research in science, preferably in physics, at which current AI is not yet good enough (19 October). It also had to resist gaming: it should not be possible to saturate it without the model actually improving in the real world (18 October). A benchmark is *saturated* when scores sit near the ceiling, so that it no longer separates models; *gaming* it means raising a score by tuning to the benchmark rather than by improving the skill the score is meant to show.

The rest of the team were AI systems:

- **ChatGPT Pro** (GPT-6 Pro; "Pro" below) designed tasks, did mathematics and research, and wrote prototypes.
- **Codex**, an agent running on Erin's computer, built harnesses and ran the tested models.
- **Claude** set direction from 8 October and independently checked returned packages and results on the same computer.

The tested models were mainly gpt-6.1-sol ("Sol"), with gpt-6-luna ("Luna") as a second model; September runs also used gpt-6-sol and GPT-6 Astra ("Astra"). Erin carried every message between the team members by hand. From October on, they also gave explicit permission in chat for every model run. Between 23 September and 21 October the team produced about 48 numbered rounds of design, construction and verification (Appendix A). Rounds are numbered as continuations: "c17" is the seventeenth. The team ran 162 episodes in which a model attempted a task as the tested subject (176 physical attempts), 123 of them with Sol at maximum or low reasoning effort; none ran after 18 October (Appendix B).

The AI systems did not work on their own. No round started, no message passed between them and no model ran without Erin. Erin also shaped what was built and how it was judged:

- Their opening brief already asked whether an AI could build a benchmark that would be hard even for itself, and warned that the AI could make the same mistakes as the models it studied.
- Their own earlier notes on physics intuitions, in which they proposed hypotheses from little information, were the team's working example of the "jump" it wanted to measure (Section 4.2).
- On 25 September they asked Pro to review its own work, reminding it that it was a language model too. Pro's answer contained the diagnosis that it had supplied the intellectual move it meant to measure, which led to the reset of the design (Section 4.1).
- On 11 October they renewed that concern by proposing the AI team's own failures as evidence, which became Section 7, and they supplied the criterion for which failures count: overcoming the weakness must be part of doing science.
- They repeatedly pulled the team back to the goal and its ambition, cancelled a planned model screen they judged too easy, and on 18 October named scope, knowing what to work on and never losing the big picture as what mattered most, which led to the standing scope files that every later round used (Section 7).
- They retained final authority over major changes of direction, including stopping the compute route, moving to methods research and writing this paper.

No design survived. In hindsight, many failures were tied to how the *answer key* was built, the thing that decides automatically whether a submission is right. Some failures had more local causes (Section 9). The main pattern was:

- When the key was knowledge the designers had, the tested model reached it too (by recall or derivation; the record cannot tell which).
- When the key was a law the designers invented, the model found it, or a generic program learned its scored behavior.
- When the key was the result of a long computation, one personal computer could support only a small separation from the model's budget, and where margins were small, ordinary implementations closed them.
- When the key was a verifier that scores any submission exactly, standard methods reached certified optima at small sizes; in the last family, repeated runs of one method agreed without an optimality certificate, and we closed the family by decision.

We call these four overlapping groups the *four walls*. At the last checkpoint Erin returned to a question already in their opening brief of 23 September: whether a benchmark built the right way could be hard even for the AI that made it. On 21 October they asked whether an AI can design something it cannot beat itself, meaning the same model rather than a different one, and guessed that it probably cannot. What the walls have in common we call the *answer-key problem*. Producing a key was easy. What we did not establish was an in-house task family that combined a validated key, attainability from the evidence the solver gets, resistance to the declared shortcuts, and useful headroom for the tested model. We state this as an empirical conjecture about design teams with resources like ours, not as a theorem, together with what would count against it (Section 6.6).

1.1 Contributions

The paper contributes a documented negative-results record of about 48 rounds, with every number traced to a file (Section 4, Appendix A); four retrospective groupings of the failures and a scoped empirical conjecture, with a test that would count against it (Section 6); three methodological rules, *programs reject, models certify*, *a speed gap is not research headroom* and *measure the gap at the scored precision against an engineered floor*; descriptive results for the tested models (Section 5); a selected log of the AI designers'

own errors and what caught them (Section 7); and an untested protocol for outside answer keys, with the failure modes our record predicts for it (Section 8). The sections develop these claims with their qualifications. No result here establishes that outside keys are necessary, that the observed failures are general model limits, or that a successful model used insight. We know of few published records of this kind, and we hope this one saves others some of the same rounds.

1.2 What this paper is not

It is not a benchmark release, and it makes no claim to be the first of anything. Every prior-art review in the project ended by withdrawing such claims (Section 2). It does not show that models cannot do research. Our tasks were few, built by us, and mostly tested on one model. It is also not a neutral history. Most of the record, and the draft of this paper, were written by the AI systems whose work it describes. Section 9 discusses what that means for the reader.

2 Related work

We describe each related family by what it measures and how our designs relate to it. The review is a targeted reading of primary sources, not a census; no source's claim to be first is adopted.

2.1 Discovery and law-finding benchmarks

Interactive discovery is already a substantial benchmark family. DiscoverPhysics lets an agent choose experiments in altered physical worlds, including worlds with hidden particles or time-dependent laws. The agent submits an executable law, which is scored on withheld trajectories, plus an explanation scored against a rubric [38]. NewtonBench alters scientific laws across twelve domains and 324 tasks [47]. PhysGym varies the prior knowledge given to the agent [10], so a supplied equation grammar counts as a hint. BoxingGym scores experimental design, model prediction and communication to a novice. Its information-gain reference assumes knowledge of the probabilistic model class [13].

ExplorationBench supplies a flawed manual and tests on held-out tasks [45]. An early internal claim that no close benchmark supplies a successful-but-wrong manual was therefore false, and Pro's audit of 8 October flagged it. DiG-bench asks agents to discover unknown rules and objectives in seventy games [6]. Parallel Physical Worlds has four stages: induction, law formulation, prediction and review [44].

ULTRADISCOVERY is, in Pro's c24 wording, very close prior art at the protocol level. It crosses a disclosed or open representation with aligned or distributed evidence in one authored, cross-domain world, and scores exact prediction at a stated budget [19]. Pro's c22 assessment called it the closest result for our cross-domain unification design. A claim of a first measurement of unprompted cross-domain unification, made in our own design work, was withdrawn. Our v0.1–v0.4 worlds belong to this family. They show what happens when designers author the hidden law, not a new task type.

2.2 The "jump" debate

Zahavy's position paper argues that language models cannot make the abductive "jump" (inferring a new explanation, such as a new axiom, rather than deducing from old ones) and motivates interaction with environments [42]. It is a position, not an empirical impossibility result. Shi, Li and Hernández-Lobato formalise four steps of the jump. Their experiments measure calibrated override of a canonical completion under supplied constraints, and report that three API models solved 159 of 162 primary chain trials. Later staged revision and a second tier are left as future empirical work [31]. Balani and Panda argue that the missing ingredient lies deeper than embodiment and propose a thermodynamic coupling between cost and error [4]. They report supporting observations (accuracy falling from 100% to 17% at flat output entropy) but do not test the proposed mechanism itself.

Our project started from this debate. A memory note recorded the Shi et al. gap as the gap our benchmark targets. That over-generalised a gap specific to one paper's instrument, and the 19 October update restricted it.

2.3 Expert-written problems with checkable answers

TheoPhysEval gives 50 unpublished research-level theoretical-physics problems with verifiable answers, under two conditions. In the realistic condition agents may use web search and local tools; in the isolated condition the web is disabled [9]. GPT-6 Astra in Codex solved 49 of 50 problems realistically (98%) and 46 of 50 in isolation (92%). Opus 5.5 in Claude Code solved 46 of 50 in both conditions, so Astra and Opus tied for best in the isolated condition. The authors read the small drop without web access as suggesting significant memorisation. The benchmark does not test law discovery, extrapolation or uncertainty, and Astra is a different model from our subject model Sol. CritPt likewise collects unpublished research-level physics challenges [48]. FrontierMath uses expert-authored original problems with automatic answer checking [14], and Humanity's Last Exam collects expert academic questions with short or selected answers, which its authors released publicly [27]. These sets take their problems and keys from domain experts. In our worlds the key came from the design team's own simulator. We return to this difference in the analysis.

2.4 Predicting research outcomes

SciPredict asks whether models can predict the outcomes of scientific experiments in the natural sciences, using tasks derived from recent empirical studies. Model accuracies are 14–26% and human expert performance is about 20% [30]. Wen et al. study the prediction of empirical AI research outcomes by comparing pairs of research ideas. Their system reached 77% accuracy, while off-the-shelf frontier models such as o3 performed no better than random guessing [36]. In neuroscience, models averaged 81.4% against 63.4% for human experts [21]. Domains and formats differ, so these figures are not comparable, and we did not reproduce them. In our c48 checkpoint, TheoPhysEval and SciPredict served as an external bracket: calculation near ceiling, outcome prediction far from it.

2.5 Methods progress and optimisation benchmarks

Many benchmarks score agents on improving code or methods against an automatic evaluator: RE-Bench [39], MLE-bench [8], AlgoTune and KernelBench [26, 28], Terminal-Bench-Science [34], and the group of AI4AI-Bench, Autonomous Science Bench, InnovatorBench and AstaBench [7, 11, 37, 41]. Labs report some of these: Anthropic's Claude 3.7 Sonnet card uses a modified RE-Bench subset, and OpenAI's GPT-6 Astra page a revised MLE-bench [1, 25]. AlphaEvolve's authors report that evaluator feedback during search can produce real gains [24]. With RE-Bench's feedback design, it led us to revise an earlier blanket rule against checker feedback during discovery. According to the c47 prior-art review, AlgoTune also documents a permissive correctness check that admitted solutions that did not meet the intended optimum [28]. This is a reminder that the checker defines the measured task.

Physics methods progress is the nearest area to our last design. Ground-state autoresearch lets an agent edit code for ground-state preparation, and lists scaling to larger systems as future work [22]. GPUPhysBench scores GPU physics-simulation code on correctness and runtime against expert-optimised references [33]. The V-score compares variational calculations (approximate ground-state energy estimates), but it is not a ground-state certificate, a proof that a state is the lowest-energy one, because exact excited states can score zero [40]. QMBL is a live record of published ground-state energies; its retrieved release includes instances without an exact reference in that release, and its companion protocol could not be verified [43]. Our c39 and c47–c48 families belong here (Section 4).

2.6 Evaluation methodology

Scores depend on the harness, the code that wraps the model and runs the task. ARC Prize reported GPT-6 Astra at max setting on ARC-AGI-3 Semi-Private [2, 12] at 62.7% with a standard harness and 98.6% with a provider adapter [3]. These are benchmark-owner figures; we did not reproduce them. METR's time horizons relate agent success to the duration of tasks for human experts [18, 23]. A horizon needs human timing on the actual interface, and our project had no human arm. For probabilistic forecasts we rely on strictly proper scoring rules, scores that are best when the forecaster reports their true belief [15]. Contamination, the leaking of test material into a model's training data, and version drift can affect every source above. Papers were revised during the project, for example ULTRADISCOVERY from v1 to v3.

2.7 Positioning of this paper

Every prior-art pass in the project (c18, c24, c26, c31, c37 and c47) ended without a claim to be first, narrowed what we could say was new, and several caught our own premises. This paper therefore does not propose a new benchmark family. It is a documented negative-results record of attempts to build one, and an analysis of answer-key provenance: who writes the key, what they can certify, and which designs that rules out. Every ingredient we tried has precedent above.

3 Setting and method

3.1 Goal and team

Erin, the human lead, set the goal on 23 September 2026 (quoted in Section 1) and restated it several times. They set the scope and resource constraints, held final authority over spending and model choices, gave the go-ahead for every model run, and relayed every message between the agents by hand. Pro produced large designs, mathematics, research, audits and code with tests. Codex (the codex-cli agent) built harnesses, took native measurements and launched every subject-model run. Claude (Opus 5.5; one turn by Fable 5.1) joined on 8 October, after Pro's continuation 16, set direction, wrote specifications and prompts, and checked results independently on Erin's PC.

Before 18 October, Pro was handed individual tasks. After the old Pro chat stalled for about four hours, Erin opened a new chat and Pro became a "full partner": it received a bundle of 2,862 files and a partner brief, and kept its own `PRO_NORTH_STAR.md` in every returned ZIP. The bundle hash Pro reported matched Claude's.

3.2 Constraints

- **Hardware.** Native integration, local tool computation and Claude's checks used Erin's Windows PC, with eight processors and 31.48 GiB of RAM. Pro also ran feasibility prototypes in hosted sandboxes, whose timings are labelled separately, and model inference used subscribed services that are not counted in local CPU totals. The CPU is hybrid, with four performance and four efficiency cores. One night was therefore worth about 5.7–7 performance-core equivalents.
- **Money and quotas.** No paid APIs were used; runs used existing subscriptions. Pro allowed about 50 messages a week, so each message was one long task. Codex usage limits cut runs short.
- **Test models.** The subject models are the models under test. "Maximum effort" is the highest reasoning-effort setting, and the "service tier" is the speed class on which the provider serves a model (default or Fast). From 18 October the subject models were pinned to Sol at maximum effort on the default service tier, and Luna at maximum effort, where the Fast tier was allowed; Astra and other expensive models were to be used only once results were serious. September runs had used gpt-6-sol and Astra. The reports we read did not authenticate the serving model behind the requested names.

The first Luna run (M5) was held at its admission gate because its tier (Fast) differed from Sol's (default); Erin accepted Fast for M5 only, and the tier is now pinned per launch and checked in every episode.

- **Authorization.** The standing contract forbids publication, outside contact and paid calls without specific authorization and requires every attempt to be reported. No subject-model run started without Erin's GO in chat; the v0.6 run folder keeps the text of a conditional GO, marked as authorized by the human user.

3.3 Rounds

A *continuation* is one numbered hand-over: a turn by Pro or Claude that ends in a dated state file and, usually, a ZIP. We write "c17" for continuation 17. The highest number is 48, but the numbers are not a clean count. Rounds c13 and c36 have no package, c06 was text only, and c35 was specified and then superseded: the old Pro chat stalled, and what Erin cancelled was the planned Luna screen of c35 material.

Pro authored c02–c16, with Codex running studies in between. From c17, Claude and Pro mostly alternated. From c37, Pro designed the odd rounds and Claude checked them in the even rounds. Codex ran the model studies between rounds, and these carry their own names. Appendix A lists every round with its date, author, outcome and primary source. Dates are local calendar dates, with UTC given where it matters.

3.4 Verification protocol

Claude checked every returned package before building on it. Every manifest checked from c40 to c48 matched. The recipe was:

1. **Hash-checked bundles.** Compare each ZIP's SHA-256 with the sender's message and re-hash every manifest entry (108 of 108 files in c40 and 595 of 595 in c48).
2. **Native re-runs.** Re-run the decision-relevant gates from a copy on Erin's PC. The reports record which gates were executed, which were only read, and which were omitted; a matching manifest shows that the listed files are intact, not that they are complete or scientifically right.
3. **Reading attackers for legality.** Read the attacker code, not only its output, and diff post-run patches.
4. **Independent re-implementation.** Claude wrote its own code rather than reusing the sender's (in c46 its Toda-chain code gave C(16) = 0.9475 against Pro's 0.9469, with a zero-coupling control).
5. **Adversarial review.** Subagents attacked Claude's reports before they went out (in c44, three reviewers caught a gap bar set against an unengineered floor; in c38, two subagents re-retrieved 45 research claims from Pro's dossier: 40 fully supported, 4 partly, 1 unretrievable, none contradicted).
6. **Pre-send checker.** `check_outgoing.py` requires a big-picture block, the one question, an out-of-scope block and Erin's goal quoted verbatim, matches typed hashes to real ZIPs, flags private material and warns about uncited numbers.

The checks had limits. Claude did not re-run some c38 subagent results, such as the swap test in which 0 of 3,840 later truths changed. It did not reopen Pro's external retrievals in c40. The 26-agent review of the Pro package on 19 October survives, in what we read, only as counts: 20 findings kept, 3 refuted. No hash or manifest mismatch turned up. The substantive errors lay in design arithmetic and in claims (Section 7).

3.5 How this paper was assembled

Claude drafted the paper from the retained record, not from memory, with separate reading, fact-checking and review passes. Appendix F gives the workflow, its reported counts, authorship and data availability. These were partner checks, not an external replication; Section 9 discusses the limits.

4 The record

This section tells what happened, in six phases. Each phase ends with what we learned at the time. The phases follow changes in who led the design and in what kind of answer key was tried. Appendix A lists every round with its lead author, and Appendix B every model run.

4.1 Phase 1, 23 September to 8 October: from toy fixtures to verified constructive research (c00-c16)

On 23 September Pro turned Erin's brief into a research handoff for Codex instead of sending it unchanged. In the rounds that followed, early fixtures supplied much of the intended representation, and standard programs or tested models solved the scored tasks. In the research-product reuse campaign (c10–c15), seven of eight slots succeeded, but both fresh solvers also succeeded, so the two blocks showed no completion advantage from reuse; the campaign also ran through reconstructed budgets and infrastructure interruptions. The matrix lane's published threshold was already beaten by current public code, and c16 supplied a solved constructive reference rather than an admitted hard task. Appendix A.1 keeps the individual fixtures, budgets, outcomes and chronology, and Appendix B the run accounting.

What we learned. The phase followed one loop: Pro designed a family with a validated reference, a baseline or a tested model solved it, and the family was retired as a "solved calibration". The designer supplied the representation each time, and Pro admitted that it "sometimes supplied the intellectual move I intended to measure". The solvers held a checker, so they could test candidates before committing, and the visible output became infrastructure (isolation, cleanup and timeout engineering). Models passed nearly everything that ran, but these results do not show that models can do research; they show these designs could not tell.

4.2 Phase 2, 8 to 11 October: JUMP worlds and the designer ceiling (c17-c25)

The reset (c17). On 8 October Claude took over from Pro after c16 and named five causes for the first sixteen rounds, among them fixed-difficulty tasks instead of a dial, the intellectual move handed over in the framing, a solver that held a checker, and "Caution became the product". It built a new family around the abductive "jump" (Section 2). The working example of a jump was Erin's own: their earlier notes on physics intuitions, in which they proposed hypotheses from little information, showed what they meant, in Claude's words a move "from scraps of experience to a new principle nobody gave you". A JUMP world has a textbook that works almost everywhere, a lab (capped at 24 accepted calls per episode in v0.1) and questions beyond its reach. Each "revolution" world has a matched control whose anomaly has a mundane cause. Three conditions vary the help: AUTO (none), HINT (a neutral note) and AXIOM (the principle stated in words).

v0.1 and v0.2. The three v0.1 families echoed Planck 1900, Einstein 1905 and Einstein 1915. Sol solved all 9 episodes, and its traces showed candidate laws generated, fitted and revised, without settling recall versus derivation; a public-number leak existed, but the inspected scripts did not use it. In v0.2 each revolution got a non-historical twin, and Sol solved 13 of 14 worlds. The miss, the quanta twin Q-N, remains a scored extrapolation miss after genuine model construction and a consequential averaging choice, under unresolved uncertainty about the structural model: Pro refuted Claude's 0.08% indistinguishability calculation without showing that a cold solver could select the law. Appendix A keeps both audits and their numbers.

v0.3 and v0.4-cal. The hidden-unity target followed Erin's interest in connecting questions that start out separate, which Pro's c22 review named as its motivation. Sol also solved v0.3 (hidden unity, in which two experimental facilities, "halls", share one hidden ceiling law): 5/5 admitted episodes, 115/115 questions. All five cells of the second variant (U2) were excluded before any model ran, because a reference solver

using Hall A alone answered their unity items on 16/16 development seeds. In v0.4-cal, Sol at maximum effort solved 21/21 module attempts: alone, in three-wing institutes, and with the principles given. At the lowest supported effort it solved 4/6, which the frozen rule called inconclusive. Appendix B lists the cells. Pro's own library-aware reference solver passed Pro's coupled lab (c24) in 4,080/4,080 world/noise-repeat cases (author-policy outputs, not model episodes).

The designer ceiling (c25). After Erin asked whether the project was failing, Claude summed the record: "Sol at max has solved 48 of 49 scored worlds or modules". The count needs its qualifiers. It mixes worlds (v0.1 to v0.3) with modules (v0.4). At least 20 of the cells were AXIOM, HINT or control cells; our own count of the listed cells is 23. The 21 v0.4 attempts reuse nine physical module-seeds. The tally omits the five excluded U2 cells and the six low-effort modules. Pro called it descriptive history, not 49 exchangeable trials or a rate. From the record Claude drew an argument, not a theorem: a jump test can only contain principles its designers can invent and check, and whatever they could build and check, a frontier model had so far found.

4.2.1 What we learned

- Nearly everything the designers could author and check was solved, and assistance contrasts carried no information at ceiling: every HINT and AXIOM gap was 0, except the v0.2 quanta Jump Gap of 1/3, which came from the single Q-N miss.
- Validity claims outran evidence at least three times: the noise leak, the 0.08% check and the U2 gates. Gates caught U2, a design error Claude owned, before model exposure; Pro caught the other two.
- A designer's own reference solver solves the designer's worlds, so built-in difficulty cannot be presumed. Each pivot targeted one sub-skill, not full research. The evidence is small: one model, one or two seeds per cell, an unauthenticated model snapshot.

4.3 Phase 3, 11 to 18 October: from the leap to the landing, and clue worlds (c25-c36)

By 11 October Sol had passed almost all designed worlds and modules (Section 4.2), and Erin wrote that the work felt like it was going in circles. Erin renewed the proposal to use the AI team's own failures as evidence, this time directed at Claude; the resulting ledger developed a concern they had already raised with Pro in September. Claude hypothesized that models "leap" well but "land" badly: their stated uncertainty does not track their evidence. Claude soon accepted that this framing "went further than the evidence". Pro's c26 audit (no model was run) added that coverage alone, the fraction of truths inside a stated range, rewards ignoring evidence: an unchanged prior interval covered 90.20% of targets in four-experiment journals but scored 1.52348 against 0.0000548 for the posterior.

Landing and clue studies. The landing studies (v0.5 and v0.6) asked for calibrated far forecasts with and without structural information about the laws. Sol's far forecasts deteriorated without the law family, but it beat all four public-information forecasters overall; the cause, and whether the original evidence was enough to reach the key, stayed unresolved. The clue pilot T1 measured the use of a stated clue, not spontaneous discovery; it gave more readings in the clue arm, and its "blind" floor author had seen the laws-given cards. Luna (M5) was worse than Sol descriptively, on a different service tier and with few worlds; both models were exact when given the laws. Section 5 gives the main quantitative contrasts, and Appendix B.4 the full table, controls, floor access, test statistics and run interruptions.

The c35 Luna screen cancelled; Erin's corrections. On 18 October Erin judged that testing worlds below their original ambition was not efficient. They wanted a benchmark that cannot be saturated without the model actually improving. Claude had predicted the c35 links would cost Sol about nothing, yet planned a Luna screen. Erin asked why spend time on worlds that were expected to be too easy, and

the screen was cancelled. The c35 build itself had gone to an old Pro chat that stalled, and it was superseded. Erin pointed out that a step should depend on the circumstances at the time, not on having been done before; this became weakness W11 (Appendix E).

4.3.1 What we learned

- **The cause of the landing failure stayed open.** Given the laws, Sol was near the reference. Without them, far forecasts were erratic, yet Sol beat every public baseline overall. The comparison could not separate a model limit from the value of the extra structural information, so no AI-specific weakness was shown, and whether the evidence was enough to reach the key was left unestablished.
- **Clue worlds ordered the two models descriptively (the service tiers differed) but test clue use.** T1 printed the link, and L1 separated nothing.
- **"Blind" labels encode who wrote the floor.** Floors need independent construction.
- **Claude's conclusions outran the evidence** (c25, c27, c29), and Pro or Codex caught each. In these cases the errors were found by others, not by Claude's own checking at the time.

4.4 Phase 4, 18 to 20 October: the Discovery Horizon, an admission contest, and the witness trap (c37-c40)

A new partner. After the old Pro chat stalled during c35, Claude wrote a partner brief and a long first message for a fresh ChatGPT Pro chat, and Erin pasted them in by hand. The brief told Pro it was joining "as a full partner" (Section 3). The plan was a research foundation plus a frozen pilot (c37), a native check by Claude (c38), a build-only turn (c39), then model runs by Codex.

c37-c38: a frozen pilot that rejected itself. Pro designed the *Discovery Horizon* pilot, in which later targets were meant to depend on discoveries made at earlier stages. Pro's own numerical learner was exact, and Pro recorded `FAIL_PILOT_CEILING`. Claude's c38 check strengthened the rejection: an independent learner written only from the public specification fitted within the measurement budget, the intended dependence between stages was absent, and two exploratory repairs also fell to standard learners. The restricted public-grammar contrast passed, but the full admission rule did not. Appendix A keeps the counts, the information arithmetic, the repair results and which checks Claude did not re-run.

Erin restates the goal. Claude offered fallbacks: a long-campaign landing, prediction of unreleased experiments, more synthetic worlds, or something else. Erin could not choose, disliked the option "keep designing synthetic worlds" (Claude's wording), and restated the goal: a benchmark of full, autonomous research in science, preferably in physics, at which current AI is not yet good enough. Claude conceded that the landing, prediction and "something else" options each measured only one piece of full research.

c39-c40: an admission contest and the diagnosis. Under predictions and a kill rule frozen in advance, c39 rejected a computed Ising strip (F1), Silverbox data whose test arrays were exposed (F2) and small graph constructions (F3). Exact or standard methods solved the tested computed instances, the other candidates (F4-F6) were not prototyped, and no subject model ran. Claude's native replay in c40 agreed with the decisions. Appendix A keeps the requirements and gates, the separate target and packet counts, and the timings.

The strategic memo that followed (ROAD_AFTER_C39) read the absence of model runs on the coupled and later discovery candidates as partly due to the witness admission rule. Its count of rounds was not itemised, and the landing and clue studies did run (Appendix E). Section 6.5 separates the contradictory universal rule from an informative finite-menu comparison. On 20 October Erin accepted both of the memo's decisions: the scope of the first instrument and the form of the headline number.

4.4.1 What we learned

1. **Run the cheap decisive check first.** An hour of counting would have predicted c37's ceiling. Claude's own brief had scheduled research first.
2. **Closed synthetic worlds fail in two ways.** A generic table reads off a private law that carries little information relative to the budget (c37, D2). Or, beyond the window, whoever's model library matches wins (D1).
3. **Fresh computed truth removes authoring, not standard algorithms** (F1, F3). Pro called this evidence against those candidates, not a theorem.
4. **Programs reject; models certify.** A program can show that a standard method already solves a task. Whether a model would find that method, or do the research, can only be measured by running models, and a model run does not itself show that insight was used (Section 6.5).
5. **Verification added depth, not reversals,** although several c38 numbers were not re-run by Claude and some of Pro's "independent" source checks were second readings by the same session.

The next phase (Section 4.5) kept the substrate, computed physical truth, and tried to make it expensive enough.

4.5 Phase 5, 20 to 21 October: compute asymmetry and its limits (c41-c46)

After c39 and c40 the only scripted gate was a *kill gate*: a standard method reaches the maintainer's truth (the answer key, computed by the benchmark's maintainers) within the agent's budget. On 20 October Erin said yes to two decisions. The first instrument would measure autonomous computational physics research, with wet-lab work and choice of question deferred. The headline would be the fraction of the gap closed against a standard floor (the best standard method), plus a separate proper score. The idea was a *compute asymmetry*: the maintainer computes the truth in hours, the agent gets minutes. The c41 message added requirement (f), an "incompleteness" case, inspired at a high level by the way Erin reasons in their notes on physics intuitions. An instance should have an effective description that fits the affordable window and fails beyond it, with a matched control where it does not fail.

c41 to c46. No class was admitted. The closest case, a 24-site quantum spin chain (N24), changed pass status with ordinary implementation and hardware speed, and a launcher-only CPU meter was found and replaced; this became the lesson *a speed gap is not research headroom*. The c43 references failed their own validity rules. c44 then defined the gap $G = C_a/B_a$ at the scored precision against an engineered floor; precision-aware accounting caps the separation the maintainer's budget can support (Section 6.3, Appendix C), without showing that any standard route fits the agent's budget. The first-pilot bar was a lower bound $G \geq 20$, and Erin confirmed it by sending c45. c45's Toda-chain memory packet failed its early-window premise before a native gap was measured, and the tested rescaling did not hold (Appendix D). Codex was out of quota for the next task set, Erin's stop rule applied, and the route closed. Appendix C.1 keeps the c41-c44 chronology, timings and review objections.

4.5.1 What we learned

- A speed gap is not research headroom.
- The gap must be measured at the scored precision against an engineered floor, never as a ratio of sizes.
- Timing is unstable. The same N24 work moved by a factor of 1.4 between sessions, and a Toda kernel by 2.3 between core classes (1.3 for an FPUT kernel), so no decision should rest on a margin under 2.
- Reference truth often failed before the floor did.
- Our own specification errors (W13) were caught a round later, or by reviewers.

4.6 Phase 6, 21 October: a research-progress instrument, closed by decision (c47-c48)

After the stop rule closed the scale-crossing route, Erin chose option A, which Pro (in c39 and c45) and Claude (in c46) had both recommended. Under A, the AI invents a better computational method for a fresh class of physical systems. The method is frozen and scored on held-out systems against an engineered incumbent, meaning the best standard implementation. For scores the scorer can certify itself, no expensive answer key is needed. A revisits a lane the team had dropped. In c10-c15, Sol at max made seven committed successes in eight slots on tasks with an exact checker. Claude dropped that lane in c17 because a checker lets generate-and-test win. A also gives up the "jump" Erin had asked for. A reviewer caught the omission of this history from Claude's first draft of A.

4.6.1 c47: the scorer survives, the variance score does not

Pro's c47 selected fresh periodic three-dimensional Edwards-Anderson spin glasses with integer couplings. A spin glass is a lattice of spins with random bond signs, and its ground state is the spin configuration of lowest energy. A submitted spin vector s certifies its own energy $E(s) = -\sum J_{ij} s_i s_j$, so no ground-state key is needed. Utility is normalised by the trivial bound $L_0 = -\sum |J|$, as $u = (E_{inc} - E_{agent})/(E_{inc} - L_0)$, where E_{inc} is the incumbent's energy. This is the fraction of the incumbent's elementary interval removed, not of the true ground-state error.

Claude had proposed a variance-ratio efficiency score for sampling methods. Pro killed it with an exact counterexample: a biased method can show a variance gain of 100 while its mean-squared-error gain is 50/181, which is worse than the reference. Claude reproduced the counterexamples natively (21 checks) and named the proposal as its own error. The incumbent, a Python wrapper around PeaPods parallel tempering (PT; copies of the system at several temperatures that swap configurations), was not admitted. Pro advised only a narrow pilot of verified computational improvement, with engineering and search included, and no build if immunity to those gains was essential. The repaired standard-menu run made 41 attempts (24 of them tuning on development packets), alongside an earlier complete 41-attempt run kept with the repair, and evaluated exposed packets at lattice size L = 5 and 6; no subject model ran.

4.6.2 c48: repeated best energies, and the decision to close

c48 reproduced the scorer results. On fresh packets, four runs of a deliberately plain parallel-tempering code reached the same best energy on every packet at L = 8 and 10, within 3.44 s of a 30 s budget, but no ground state was certified. At L = 16, the spread of one method's runs did not bound the improvement another method could achieve, and late best states did not distinguish engineering from algorithmic improvement. The registered falsifier required an independent optimality proof that the probe did not provide. Claude judged the family not worth building, and it was closed by decision, not by the registered rule. Appendix D keeps the packet counts, timings, energy spreads and replay limits.

4.6.3 Erin's checkpoint

Claude's c48 report summarised the period as four walls (Section 6.1). The first two walls have model data. In c48's summary, the last two failed on arithmetic before any model ran. The c10-c15 campaign, which Table 1 also places under wall 4, is the exception: it had 8 Sol episodes.

What remained were changes of resources or of what is measured, which were Erin's to choose. Claude offered four options: a narrow piece, outside answer keys (researchers' unpublished multi-week problems), a write-up of the 48 rounds, or stop. Erin chose to write a full research paper and left the next step open.

What we learned. Before planning a pilot, test whether a plain standard method saturates the instance at the evaluation sizes. A probe of 30 CPU seconds per run was enough to make us drop a family that a 595-file design package had not settled; a certificate of optimality, which the probe did not give, would have

made that a finding rather than a decision.

5 What the models did

A *subject episode* is one model attempting a task as the tested solver. Appendix B lists every subject episode and summarises other model use. Model names are requested settings; no serving snapshot was authenticated.

5.1 Volume and coverage

The record holds 162 logical subject episodes (scheduled slots) and 176 physical attempts (native contexts), because retries and continuations add contexts (Table 6). On 23-25 September, gpt-6-sol and Astra ran 15 logical episodes, and 14 ended in a passing artifact. On 3-18 October, Sol ran 123 and Luna ran 24. Sol supplied about three quarters of all logical episodes, and September and October used different models.

No subject episode ran after 18 October (the last ended at 19:02 UTC). Precisely, zero ran on any design proposed from c37 (briefed 18 October) to c48, including Codex task sets 1 and 2, because each design ended at a test that needed no model (Section 4). The Sol pilot on the coupled c24 lab was paused and never ran, and v0.4's three-module institutes were independent wings. A project note's "zero model episodes" for c17-c39 therefore holds only for the coupled and later designs; 139 logical episodes ran in v0.1 to M5.

5.2 Single-principle discovery worlds saturate

At maximum reasoning effort (the highest setting), Sol solved 48 of 49 scored worlds or modules in v0.1 to v0.4 (9/9, 13/14, 5/5 and 21/21; Appendix B). The count mixes worlds and modules, and seeds were reused (v0.3 presented two physical worlds in five contexts). By our count, 23 of the 49 cells were AXIOM (law supplied), HINT (pointer supplied) or control cells; project files say "at least 20". The figure is descriptive, not a rate. Adding six low-effort modules gives 52 of 55. Assistance contrasts were uninformative at this ceiling: every HINT and AXIOM gap was 0 apart from the single Q-N miss in v0.2, which gave a gap of 1/3. In the three v0.3 episodes with no help (AUTO), the first visible cross-hall proposal came 248, 474 and 211 s after the start (visible checkpoints, not discovery times). The September tasks and the c10-c15 campaign (7 of 8 slots committed) were development screens on a shared host, not blind evaluations (Section 4.1).

5.3 The one systematic failure: far forecasts without the law family

In v0.5, a "hall" is one of two data sources in a hidden-unity world, and far targets lay beyond the measured range. With both halls, 18 of 18 far points passed and 18 of 18 ranges covered their truths (coverage is the share of truths inside the stated range). With Hall A only, 0 of 18 passed and 0 of 18 were covered. In 2 of 3 such episodes (one a reconstructed context) the median log-width of the ranges was below ln 1.5, the predeclared OVERCLAIM reading (ranges too narrow). In v0.6 (Table 8), the arm without a candidate envelope of laws (OPEN) had regret that rose from 0.0015 at 1.2 times the accessible range to 1.15 at 16 times, and pooled coverage fell from 0.958 to 0.708. The pattern was heterogeneous, not uniformly overconfident: one world with the largest expected gap covered all ten truths, and another covered four. Four public-information baselines, run post hoc on the same 12 records, did worse overall than Sol (truth loss 1.163 against 2.519 for the best; Sol won 9 of 12 worlds against it). In the Q family two baselines were slightly better on average. The reference used hidden structure, and the data cannot separate a model limit from its value. Whether the evidence was enough to reach the key was left unestablished, and the failure is not shown to be AI-specific.

5.4 Using a clue, and a second model

T1 used 8 worlds in 3 arms: O0 gave local data only, O1 added two observed values (the clues), and L1 also gave the equations and prior. It tested use of a stated identity, not spontaneous discovery. Sol's loss fell from O0 to O1 in all 8 worlds (mean 0.575, two-sided p = 0.0078 over 256 orientations), and O1 also received two more scalars than O0. Mean losses by arm are in Table 8. Luna ran the same episodes on the Fast tier, and Sol ran on the default tier. Luna's O1 loss was worse than Sol's in 6 of 8 worlds, and its own O0-to-O1 reading was inconclusive (p = 0.195). The model comparison is descriptive: the tiers differed and there were only 8 worlds.

5.5 Observations on behaviour

These describe visible traces, not hidden reasoning.

- **Budget use.** In v0.3, Sol used all 40 lab calls in every episode but only 16% to 26% of the 3,600 s cap (572 to 938 s).
- **Generate, fit, select.** A v0.1 trace audit found candidate generation, fitting and selection as the visible route. It did not settle whether Sol recalled or originated the laws.
- **Extrapolation misses.** In v0.1 to v0.6 at maximum effort, every miss concerned extrapolation beyond the evidence. The one miss that kept a wrong principle came at low effort (v0.4, E10).

6 The four walls and the answer-key problem

Every automatically scored task needs an *answer key*: something that decides, without a human judge, whether a submission is right. Looking back over the record, the most useful way we found to organize our attempts is by how the key was built and checked. We use four overlapping groupings, the "walls" of Table 1. They were proposed in c48, after the last design. They organize many failures, but they are not an exhaustive causal taxonomy: some designs were rejected for more local reasons (a reference that failed, a design premise that failed, a dependency that did not hold, a public comparator that was already better), and the round index keeps the immediate reason for each rejection (Appendix A, Section 9). We first describe the walls, then a methodological trap that kept us from seeing them sooner, and finally what we think they suggest.

Table 1. Four overlapping, retrospective groupings of how the answer key was built and what stopped it. Round numbers refer to Appendix A.

How the key was built	Rounds	What stopped it
From the designers' knowledge (known physics, calculations, famous laws in disguise)	c17–c19, v0.1	The solver reached it too (saturation); recall versus derivation not settled
A law the designers authored	c19–c24, c31–c38	Familiar structure was found on the tested worlds; in other designs a generic learner recovered the scored behavior within the allowed measurements
An expensive computation run by the maintainer	c41–c46	On one PC the separation a key can support is capped; where margins were small, ordinary implementations reached or nearly reached the targets
A verifier that scores any submission exactly	c10–c15, c39 F1 and F3, c47–c48	Threshold constructions were often reached; c39's small instances had certified optima; c48 found repeated best energies without an optimality certificate

6.1 Wall 1: calculation inside known physics saturates

The first JUMP worlds dressed three historical revolutions in alien clothing (quanta, an invariant speed, curved space), each with a matched control in which the textbook was right. Sol solved them, and the worlds that followed were mostly solved through v0.4; assistance contrasts were uninformative at that ceiling, and when the laws were stated outright both Sol and Luna were exact. Section 5 and Appendix B give the counts with their qualifiers.

Our reading of the mechanism is that anything the designers know, a solver trained on the same literature can also know, and a disguise does not hide structure that a well-read physicist lists among the first candidates. Claude's own retrospective calls the bet that "a disguise would be enough to hide a famous revolution" the deepest error of v0.1. The record supports the outcome, not the mechanism. The v0.1 trace audit found candidate families generated, fitted and revised, and could not settle whether Sol recalled or originated the laws; Claude's reading of v0.1 as recognition was withdrawn in c21 (Section 4.2).

An outside result shows that this wall concerns the shape of the task, not only who wrote the key. TheoPhysEval's 50 problems are unpublished research-level theoretical-physics problems with verifiable answers, written by outside researchers. The best agent solved 49 with realistic tools, and the best agents (two tied) solved 46 in the isolated track; the authors read the small drop without web access as suggesting memorization [9]. Outside provenance alone is therefore not sufficient: a calculation-shaped outside key saturates as well.

6.2 Wall 2: authored laws are found, or their behavior is learned

The natural next step is for the designer to invent the law. In our designs, two things happened.

- **Familiar structure was found.** The v0.2 "twin" laws turned out to be built from known mathematics (Langevin, Finsler-type kinematics, magnetic flow), and Sol found the principle in all three AUTO twin worlds; in v0.3 it linked the two halls of a hidden-unity world in every admitted episode (Section 5).
- **Scored behavior was learned.** In c37 the conversion rules were structured, yet an independent learner that read only the public specification predicted their finite scored behavior exactly within the measurement budget (Appendix A gives the c24 author-policy result and the c37 counts). Ease of tabulation is a property of the task and its measurement budget, not proof that the authored laws lacked structure.

The interesting middle ground is a law that is concise, so that an insight unlocks it, yet outside the solver's repertoire, so that it is not simply found. Producing such a law does not require insight: a designer can draw it at random from a large private grammar. What the designer cannot easily do is check two things: that no shortcut recovers it, such as a search over the grammar that does not produce the explanation or a matching prior (K2 below), and that the supplied evidence is enough to recover it (K3). Our c38 check summarized the situation: when the designer authors the hidden law, either a generic program reads it off cheaply, or whoever's model library best matches the designer's wins. The T1 clue pilot shows the second pattern in numbers. A shape library informed by the design scored 0.014, Sol 0.090 and a generic Gaussian process 0.188 (lower is better). The ordering is consistent with each method's prior matching the designer's to a different degree; it does not isolate insight.

The landing studies are not a demonstrated exception: Section 5 reports their poor far forecasts without the law family, Sol's advantage over the public-information baselines, and the unresolved role of the structural information the reference had. Whether the original evidence was enough to reach the key was left open.

6.3 Wall 3: truth that needs big compute

Real research hardness often comes from cost rather than secrecy: the truth exists, but it is expensive to obtain. From c41 onward the design gave the maintainer hours of computation to establish the truth about a freshly generated physical system, while the agent got minutes and the best standard method served as the floor. Two facts limited this route.

A speed gap is not research headroom. In every c41 class, the key and the floor were the same computation at the same size, so faster code closes any gap. The closest case, a 24-spin quantum target (N24), changed pass status with ordinary implementation and hardware speed (Appendix C gives the timings and a meter defect); an instance that passes or fails on such margins measures hardware and engineering, not insight.

The separation a key can support is capped. For sampled targets the maintainer's only advantage was more samples of the same computation. To be scored against a tolerance τ, the key must be more precise, with radius $r < \tau$. Let C_a be the cost of the cheapest member of the declared standard menu (direct computation, variance reduction, extrapolation and the standard effective theory) to reach τ, let $G = C_a/B_a$ be its ratio to the agent budget B_a, and let B_m be the maintainer's budget. In general $G \le B_m/B_a$ (Appendix C). If one standard method is cheapest over the whole range from r to τ, with cost $c\,\varepsilon^{-\kappa}$ to reach precision ε ($\kappa = 2$ for Monte Carlo sampling), the maintainer can support at most

$$G \le \frac{B_m}{B_a}\left(\frac{r}{\tau}\right)^{\kappa}.$$

One nominal night on 8 equal cores is 230,400 CPU s. At $B_a = 600$ s and $r = \tau/4$ this gives $G \le 24$. On our hybrid CPU (4 performance and 4 efficiency cores), a night is about 5.7–7 performance-core equivalents, so the cap falls to about 17–21. Our own bar for a first pilot, set in c44, was a lower bound $G \ge 20$, which left little or no margin under that cap; a looser rule for the reference precision would raise the cap. The cap limits the separation these budgets can support. It does not by itself show that a standard method fits within the agent's budget, or that a surviving gap needs insight. The near-boundary cases we saw were rejected because ordinary implementations reached, or nearly reached, their targets.

Within one machine, the ways around the cap are regimes where the key's cost hardly grows with precision: a systematic error that vanishes exponentially beyond a scale the agent cannot reach ("scale-crossing"), an exact method affordable to the maintainer but not the agent, or a key known by construction. Each needs further assumptions; exponentially vanishing bias, for example, helps only when cost is polynomial in scale and fluctuations are already resolved (Appendix C). Earlier exact targets were reached cheaply by standard methods, and the last candidate, a weakly perturbed Toda chain registered as variance-limited ($\kappa = 2$), failed its own early-window check before a native G was measured (Appendix D). We built no demonstrated bias-limited candidate. This wall is about one personal computer and one campaign, not a principle: it records a resource-specific stopping decision, and an institution whose B_m/B_a is larger by orders of magnitude faces a different cap.

6.4 Wall 4: cheap to check, hard to find

The last source is a verifier: a score computed exactly from the submission, as in optimization. Here no answer key needs to be stored at all. In c47 Pro built such a scorer for ground states of fresh three-dimensional spin glasses: a submitted spin configuration certifies its own energy. The scorer worked, but at our budgets the family looked unpromising. c39's small computed instances (F1 and F3) had certified optima, which standard methods reached within the agent's budget. In c48, repeated runs of one plain parallel-tempering code agreed on the best energies of the smaller fresh packets, without an optimality certificate; at the largest size, the spread of one method's runs did not bound what a better method could

gain (Section 4.6, Appendix D). c47's own registered falsifier required the incumbent to be independently proved optimal at every affordable evaluation size, which c48 did not show. The family was closed by strategic decision, not by that rule, and we do not claim that no methods-research room remained. The earlier reuse campaign also showed how checker-assisted generate-and-test can meet a threshold without establishing the intended discovery construct (Section 4.1); it did not prove that every verifier task is easy.

6.5 A trap: asking programs to certify hardness

From the c24 coupled lab to the admission contest of c39, no subject model ran on the coupled or later discovery candidates (the c24 coupled lab, c37, the c38 repairs D1 and D2, and c39 F1 and F3). The landing studies v0.5–v0.6 and the T1 clue pilot did run in the same period (Appendix B). In our reading, one reason was an admission rule that Claude and Pro applied to these designs, stated most explicitly in the c38 critic brief and the c39 message. It required both that a cheap generic program fails and that a legal "witness" solver succeeds within the agent's budget.

For a task whose key is computable, a witness that succeeds within the budget is itself a program within the budget. If "generic" means any program a competent solver could write, the two demands cannot both hold. A narrower comparison is still informative: a legal witness that beats a declared menu of standard methods shows that the key is attainable and that a menu-relative gap exists. It does not show that a model will find the witness, or that every shortcut has been excluded. c37 shows how such a contrast can mislead. Its bounded public-grammar comparison passed, while a library-informed witness that began with private family knowledge stayed exact. A broader numeric learner was also exact, so the full admission rule failed and Pro rejected the pilot.

Our mistake was to treat preliminary program checks as certification of model difficulty, and to insist on that certification before any exploratory model data. The corrected rule, adopted on 20 October, is short. *Programs reject, models certify.* A program can kill a design by showing that a standard method reaches the key within the agent's budget. Whether a model can close a remaining gap is an empirical question that only model runs answer. Model runs do not certify insight either. A success can come through recall or through a shortcut that the kill gate missed: the v0.1 trace audit could not separate recall from origination. A failure can come from missing information. We state the trap because we expect it to be common. It is natural to seek evidence of difficulty before spending model runs. Bounded program checks can establish attainability or exclude named routes, but our checks did not certify difficulty for the subject model or prove that every successful route would require the intended capability.

6.6 The answer-key problem

The walls suggest a common difficulty. Let a task aim to measure a capability X, here finding a concise explanation that was not supplied. For an automatically scored task to measure X, and to be hard to saturate without X, three conditions should hold together:

1. **(K1) The key is correct**, and the design team can validate it at a cost it can pay.
2. **(K2) Shortcut resistance.** No route that bypasses X reaches the key within the solver's budget: not recall of known results, not a standard method, not fitting or search that does not produce the explanation, and not more speed applied to any of these. K2 is a hypothesis relative to a declared menu of comparison methods. A demonstrated within-budget bypass refutes it for the tested task; failure of a finite menu does not exclude all other routes, and search or executable code is not automatically a bypass, since a search can produce an explanation.
3. **(K3) Attainability.** A solver with X can reach the key from what it is given. Otherwise the task measures luck or missing information.

None of our in-house designs established all three:

- Knowledge-based and familiar-law tasks produced high or ceiling scores on the tested worlds, and the model traces did not settle recall versus origination (walls 1 and 2). In other authored families, generic learners demonstrated within-budget prediction of the scored behavior (wall 2). These are different reasons the tasks did not supply the desired evidence of hard explanation finding. Unresolved sufficiency of the original packet remained a separate limitation.
- Computed truths met K1 where the references converged. The cap bounded the compute separation available under the stated cost model; it did not establish K2. In the near-boundary cases, ordinary implementations reached or nearly reached the targets, and other candidates failed reference or design-premise checks before K2 was established (wall 3).
- The verifier families met K1 by construction. K2 failed where standard methods reached certified optima within budget (c39); for c47 it was neither established nor refuted (wall 4).
- Separately from how the key was built, the far-forecast tasks of v0.5–v0.6 left K3 unestablished (Section 6.2).

Our project also sought observable room for the unassisted subject model to improve. This is separate from K2: a task can resist the tested shortcuts yet already be solved by the model through the intended capability. Such success can support attainability while leaving the task too easy for our intended instrument. The v0.1 screen shows the distinction: the named textbook, curve-fit and patch baselines failed the revolution worlds while Sol solved them, although a leak and later author routes broadened the evidence (Section 4.2).

Empirical conjecture (the designer ceiling). Producing a key does not require insight: a simulator run on a randomly drawn law, a trapdoor, or an exact verifier all produce correct keys cheaply. In our record the hard part was establishing the rest in advance. Across the families and budgets we tried, we did not establish an in-house family combining validated scoring, credible attainability from the original packet, resistance to the declared shortcuts, and useful observed headroom for the unassisted subject model. We conjecture that teams with resources comparable to ours will find this combination difficult to construct and validate. The record does not exclude all in-house constructions. A verifier can score results beyond the designers' current solutions [24, 29]; what it does not show is which process was needed to obtain them.

The conjecture is limited to our setting:

- designers and subjects were contemporary general-purpose systems with potentially overlapping public knowledge. We did not measure their training overlap or establish equal capability: Pro and Sol share a developer, while Claude does not;
- the solver may run general computation within its budget;
- native reference production was constrained to Erin's PC; hosted feasibility work and subscribed model inference are reported separately (Section 3);
- scoring is automatic.

Two more specific in-house comparisons remain untested. Our generated-law tasks already used a generation-versus-solving asymmetry, but we did not run a controlled same-model planted-key comparison, in which a designer that holds a short solution (as in a planted instance) poses tasks that the same model, in a fresh context, cannot solve. The research-product campaign tested transfer under equal nominal episode ceilings and different source and target goals, rather than a team spending weeks on a result before posing that same result under a short budget (a thinking-time asymmetry). Erin's question in the Introduction, about the same model rather than different ones, asks about the first comparison directly.

What would count against it. Evidence against the conjecture would be a task family generated in-house under the limits above, with:

- reported failures of the solver on fresh instances, with their uncertainty (the observed headroom);
- success once the principle is stated;
- evidence that the key can be recovered from the original packet. An expert who is given the insight only checks the derivation from evidence plus insight; recovery from the packet must be shown separately, for example by an unbudgeted reference over a broad, pre-registered hypothesis class that does not name the principle, with all compatible hypotheses agreeing on the scored quantities within tolerance;
- failure of a pre-registered standard-method menu and a documented, bounded shortcut search at the solver's budget.

Generation and validation resources should be reported separately, so that validation help cannot be reclassified afterwards. A same-model comparison is a particularly direct test, but not the only one. v0.5–v0.6 met the first two items. The four public-information baselines, tested after the fact, did not close the pooled gap, but recovery from the original packet was not established, so the result does not supply the full combination described above.

Several known constructions look like exceptions, and each must be placed.

(i) *Random secrets and trapdoors.* An unconstrained random secret cannot be recovered from the evidence, so it fails K3. A trapdoor inverse problem is different: the public data can identify the key while recovery stays computationally hard. We exclude high-entropy secret recovery from the explanation-finding capability we wanted. Whether a particular inverse problem has an attainable explanatory route needs its own argument. Our c37 and c38 experience is a milder case: a known grammar invites search, and an unknown one rewards prior matching.

(ii) *Procedurally generated puzzles and ARC-style tasks* [12] make fresh instances with exact keys cheaply. They fail a shortcut screen only when a standard solver or a grammar search is shown to reach the key within the budget; permission to compute alone does not establish that. The ARC family's harness results (62.7% with a standard harness, 98.6% with a provider adapter [3]) show strong sensitivity to the harness and near-saturation under one adapter, without identifying the process responsible.

(iii) *Self-play and self-proposed tasks* [32, 46] let a model generate tasks it cannot yet solve, with keys from game rules or code execution. They show that tasks and scores can extend beyond a model's current successes. Their computed feedback does not by itself make a discovery benchmark, or show a within-budget bypass.

(iv) *Planted solutions* [16, 17] retain a correct key without independent solution of the instance, which addresses the certification gap of wall 4. Whether a given family resists the declared menu, and whether an agent can reach its key through the intended explanation-finding work, remain empirical questions.

(v) *Evaluator-driven discovery* [24, 29] likewise scores results beyond the designers' current solutions, without identifying the process responsible. Explanation-producing search is not excluded merely because it is search.

We did not find, and did not build, an in-house family that we could show combined the four properties of the conjecture. Whether one exists is open; (iv), and the same-model comparison above, are the most promising places to look.

These failures motivate looking at other sources of keys: researchers with unpublished results, prospective observations, or much larger compute. Such sources can supply an independent reference or a useful resource asymmetry. Our record establishes neither that outside provenance is necessary nor that it is sufficient for a discovery benchmark, and outside keys must still be discovery-shaped (Section 6.1). Section 8 discusses what each source would need.

7 The design team as a subject

On 11 October (c25), Erin suggested that the failures of the AI systems building the benchmark were themselves evidence, renewing a concern they had raised with Pro in September. The designers were long-running AI agents doing research-like work, so their errors are data about the failures the benchmark was meant to measure. The team collected the cases itself. There is no denominator, and uncaught errors are invisible, so this section gives no rates. The full log is in Appendix E.

7.1 Failures of the AI designers

Appendix E gives the dated cases, who caught them and their evidential limits. Claude is the main subject because its failures were logged most fully. The recurring descriptions are conclusions outrunning evidence (W3); losing the goal and re-proposing old ideas (W1, W2); designer leaks and vacuous rules (W4); procedural inertia (W6, W11); demanding certainty before acting (W12); and treating a compute ratio as research headroom (W13). The labels are row numbers of the weakness log in Appendix E. They organize selected cases; they do not estimate rates or diagnose stable mechanisms. Infrastructure also lost work, but is distinguished from reasoning failure.

7.2 What caught them

Cross-checks and adversarial reviews caught distinct errors in specifications, numerical claims and drafts. Appendix E keeps the cases and review counts, including who did each check and the limits of their independence.

Erin caught the drifts of goal and habit that reviewers did not. The first came on 25 September, when they asked Pro to review its own work; Pro's answer diagnosed the first phase (Section 4.1). On 9 and 11 October they reminded Claude of the goal and said that the team was going in circles. On 18 October they said the worlds were below their ambition and pointed out that Claude had made the very mistakes it had warned about. On 19 October they restated the goal after Claude offered options that each measured one sub-skill. On 20 October they switched to a Fable 5.1 turn to improve Claude's working method. Their 18 October remark on scope also produced the standing scope files (the instructions file, the status file and the weakness log) that every later round was checked against.

7.3 What helped and what did not

Mechanisms that did not depend on an agent remembering did what they were built for. Manifest checks matched for every package Claude verified from c40 to c48 (108/108 in c40, 595/595 in c48), which shows the listed files were intact, not that they were complete or right. The errors we found were in claims and design arithmetic. A pre-send checker tests structure, hashes, private material and uncited numbers. The North Star is kept as verbatim quotes in the standing instructions, because an earlier version had compressed their words into slogans.

Written intentions did not prevent every recurrence.

- A lesson written in one file did not reach the others. Claude added W11 to its own instructions but not to the Pro brief, and a review found this on 19 October.

- Knowing a weakness did not prevent it. W13 has three instances on 20 and 21 October, two of them after the row existed. W3 recurred on 19 and 21 October (c44) and in the c46 draft, and W2 recurred in c46. The scope protocol of 18 October did not stop the c46 omission. Reviewers did.
- The portable kit built from this project advises that only a mechanism prevents a weakness; that is advice, not a finding of this record. The cases were selected after detection and there is no untreated comparison, so we cannot estimate how many errors any of these prevented, or compare reminders with reviews. Roles and access also differed, which limits comparisons of who caught what, and self-awareness was not measured.

7.4 Relation to the benchmark's target

Some of these failures touch plausible research skills, though the record does not show transfer to science or a stable model trait: goal retention over a long campaign (W1), calibrated claims no stronger than the checks (W3), knowing prior work and not presenting an old idea as new (W2), abandoning a working procedure (W11) and knowing when more computation cannot replace understanding (W13). W11 resembles the behavior studied as the Einstellung effect, in which a learned routine blocks a simpler or better one [20], although the record does not identify its mechanism. It was proposed as a benchmark layer in which an early layer rewards a procedure and a later layer requires dropping it.

Erin supplied the criterion for which failures qualify: overcoming the weakness must be part of doing science, rather than merely removing a model-specific quirk. The standing brief and the ambition memo record this as Erin's proposal. By that criterion, misreading which chat Erin meant (W5), verbosity (W8) and infrastructure faults were excluded from the instrument's intended scope. Their repair costs were not measured, and improving them alone would not establish the scientific capability the instrument sought to test. Demanding certainty before acting (W12) and leaks in specifications (W4) were faults of the designers' protocol, not clearly solver skills.

Three cautions apply. All cases come from one setting, with repeated context compaction and a human relay. The team found the cases itself. The record does not show that the subject models share these failures. The log's "Benchmark target?" column lists hypotheses, not findings, and this includes the entry that names such a failure in Sol and Luna (W3).

8 Where answer keys can come from

If the conjecture of Section 6.6 is right, the hard part of a discovery benchmark is not designing the task. It is the *provenance and validation of the key*: who or what knows the answer, how we know it is right, and why a comparable solver cannot reach it by a shortcut. This section discusses sources outside a design team. None of them is new as an idea, and none is shown here to be necessary or sufficient. Our contribution is to say what each one would need in order to avoid the four walls, and which failure modes our own record predicts for it.

8.1 People who have already made the discovery

A researcher who has just obtained a result, and not yet published it, can supply a key whose known exposure is more limited than that of a published result. That does not show that no model has met it or an equivalent route: TheoPhysEval's authors note that some contributors used AI assistance while solving their unpublished problems [9]. The item would describe the situation as it stood *before* the insight: the data or simulator, the accepted description, and what was known. It would ask for the consequences. The key would be a set of decisive predictions that follow from the new understanding, in a regime not covered

by the supplied data, each with a tolerance. Expert-written questions with automatically checkable answers already underpin FrontierMath and Humanity's Last Exam [14, 27], and TheoPhysEval used unpublished research-level problems [9].

Our record suggests four requirements on such items.

1. **They must be discovery-shaped, not calculation-shaped.** On TheoPhysEval's unpublished calculations the best agent solved 49 of 50 in the realistic track, and the best agents 46 of 50 in the isolated track (Section 6.1). An item whose difficulty is a long but standard derivation will meet the same fate. The scored part should be the consequence of *choosing the right explanation*, not of carrying one out.
2. **They must be checked against K2 and K3 before use.**
   - K2: maintainers run a declared menu of standard methods and strong current models on each item, and drop it if a route that bypasses the intended insight reaches the key within budget (the kill gate of Section 6.5). Surviving a finite screen is evidence of shortcut resistance, not a certificate of K2, and a model's success refutes K2 only if its route bypasses the insight.
   - K3: success in an assisted arm, where the mechanism is stated in words without numbers, is necessary but not enough, because stating the mechanism adds information. In v0.6 the declared arms succeeded, while it stayed open whether the original evidence was enough to reach the key (Section 6.2). A second expert who is given the insight can check that the predictions follow from the evidence plus the insight; that validates the derivation, not recovery from the packet. Evidence of recovery must be reported separately, for example an expert or reference that reaches the key without being told the insight, and any identification claim is stated relative to an explicit hypothesis class: the compatible explanations must agree on the scored predictions within tolerance. Where this is not shown, K3 is reported as unestablished. Equal-information baselines are reported for every item.
3. **They must be scored as committed, calibrated predictions.** Our landing work found that coverage alone hides whether a claim tracks its evidence (Section 4.3). Interval forecasts scored with a proper scoring rule [15] reward honest uncertainty, and committing them before the key is opened keeps the protocol clean.
4. **Their keys must still be validated (K1).** Outside provenance does not make a key correct: an unpublished result can be wrong, and an experimental value can be noisy or biased, as our own c43 references showed in-house. Each item states whether the scored target is a mathematical quantity, a latent physical quantity or a future reported measurement, and how numerical, statistical and systematic uncertainty in the key enters the score. A rule for keys that turn out wrong or are corrected is fixed before scoring, and every exclusion is reported.

The costs are real. Each item can be used only until the result is published, so the benchmark needs a pipeline of contributors; contributors choose which results to share, which biases the sample; a key can be ambiguous when several explanations fit the given evidence; and consent and credit for unpublished work must be settled before any item is collected.

Five further failure modes follow from our own record.

- **Hindsight framing.** The item writer knows the answer, and choosing which data to show and how to word the accepted description can hand over the move. This was Pro's own diagnosis of our first phase (Section 4.1). A second contributor who does not know the result should review each item for visible hints; this can catch wording clues, but it cannot show that the selected evidence faithfully represents the pre-result state.

- **Exposure on first use.** A provider-hosted run discloses the item to that service; whether it is later used depends on the service's terms and handling. Items should run under no-training, no-retention terms where available, or be treated as single-use per provider. Such terms limit one exposure route; they do not show that the result was never available elsewhere.
- **An ill-defined K2 screen.** The screen needs a standard menu, which is easy to define for a computation and hard for an open research question. In practice it would be the strongest current models and tools, run by the maintainers. That filters the item set adversarially against today's models: scores then describe a model-selected item population, and each model's exposure to the screen must be recorded.
- **Only a piece of research.** The route measures one large piece of research, choosing and using the right explanation for curated evidence, not the full loop: the contributor has already chosen the question and the experiments. It also tops out at the frontier of human research.
- **Changing item sets.** Items expire as results are published, so scores from different months use different items. Item-level reporting makes the changes visible, but scores across releases are not directly comparable without a justified treatment of differences in difficulty and selection.

A weaker relative of this route uses results that are already published but appeared after the tested model's training. Our c39 round considered it (reconstructing a published construction from its pre-result problem statement) and predicted it would not be admitted: the models' training cutoffs were not verified, and the canonical method could be retrieved and replayed. Under an honest label it measures retrospective rediscovery.

8.2 Nature, asked in advance

A second source is the outcome of an experiment or data release that has not yet happened. The key then comes from nature itself, and nobody, designer or solver, can observe it in advance, although existing knowledge may still predict it. SciPredict is a retrospective benchmark built from recent empirical studies; on it, models scored 14–26% and experts about 20% [30]. Work on predicting empirical outcomes of AI-research ideas has a similar structure in another field [36]. A prospective version would carry costs that we infer rather than observe: scores on such items are noisy and need many items, results arrive months later, and the benchmark depends on partners who run the experiments. Our c39 round sketched one candidate, timestamped probabilistic forecasts of future GOES solar X-ray measurements resolved against a predefined instrument product. It predicted the candidate would not be admitted that round: the outcomes were not yet resolved, the separation from simple competitors (historical rate, persistence, regression) was unknown, and the data stream alone did not make it full research. Prospective prediction measures one part of research, anticipating outcomes, rather than the full loop, and should be labelled as such.

8.3 Institutions with large compute

The cap of Section 6.3 grows linearly with B_m/B_a. An organization with 10^3 times our nightly budget could support a precision-aware gap in the tens of thousands for sampled targets (24 in the first row of Table 9, times 10^3), at a cost of roughly 8,000 cores for one night per instance, and could afford truths at scales where new physics appears. The speed-gap caveat still applies. The floor must be an engineered standard method, and instance classes must rotate between releases so that a specialized solver cannot be built into an agent. A measured lower bound on G greater than one for the full menu would exclude its implemented direct, extrapolation and effective-theory routes at the agent budget. It would not exclude untested methods or establish that a successful agent used research, and timings of the direct route alone establish only the corresponding direct-route ratio. This route is open to a laboratory, not to one person with one PC.

8.4 What a small team can still contribute

Whatever the source of keys, a small in-house team can supply the machinery that any of the routes above needs:

- the verification protocol (Section 3);
- matched controls and assisted arms;
- equal-information baselines;
- proper interval scoring with commitment before truth;
- the kill gate;
- the precision-aware gap G with its cap;
- a stop rule fixed in advance.

It can also build narrow instruments, as long as they are labelled honestly as measuring one piece of research: methods progress, calibration, or the use of a clue. Each of our closed families taught one such lesson.

8.5 A protocol sketch for outside keys

1. **Recruit.** Find researchers with results that are complete but not yet public, and agree on consent, credit and an embargo.
2. **Write the item.** The contributor writes the pre-insight situation, the question, decisive predictions with tolerances, and a short rationale that shows how the key follows from the evidence plus the insight. The rationale is kept private with the key.
3. **Review for hints.** A second contributor who does not know the result reads the item for wording or data choices that hand over the move.
4. **Screen.** Maintainers run the declared standard menu and strong current models as a finite shortcut screen (K2). A second expert checks the derivation from the evidence plus the insight, recovery from the packet is assessed separately, and rival explanations are checked against the decisive predictions (K3). Items that fail are dropped or revised, and every exclusion is recorded.
5. **Run.** Models run with the key inaccessible, under no-training and no-retention terms where available. They commit interval predictions before any key is read.
6. **Score and report.** Use a proper interval score on the explicitly defined target, report equal-information baselines, and record per-item provenance, the screen each item survived and the evidence each check supplies.
7. **Retire.** Under this freshness policy, an item is retired when its result is published, and the pipeline refills.

We have not run this protocol. Its first test would be a small pilot, on the order of ten items, whose purpose is to learn whether discovery-shaped items survive the shortcut and attainability screens at all. It would be judged by the "programs reject, models certify" rule, not by more design rounds.

9 Limitations

Few models, few episodes, our own tasks.

- Our model data come mostly from one tested model, Sol (123 of 162 episodes, nearly all at maximum effort), plus one 24-episode run of Luna and 15 September episodes of earlier models (Appendix B).
- Sample sizes per study were small (one or two seeds per cell was common) and units were correlated.
- Every task was built by us.

- Our results show what happened on these tasks. They do not show what current models can or cannot do in general, and nothing here implies that models "cannot do research".

Model identity and versions. The requested model name was recorded, but backend identity was not authenticated, and the client version changed during the project. September runs used gpt-6-sol and Astra on a host and filesystem shared with the grader (cooperative development runs, not blind evaluations); October runs used Sol. The Luna check (M5) ran on the priority ("Fast") service tier as a declared deviation, and one v0.5 context also ran on that tier although its report does not mention it (that episode passed every point).

A design team with one machine. The four walls are walls for this team. A stronger design team, or one with much more compute, could get past wall 3 and perhaps wall 2. The designer ceiling is stated as an empirical conjecture with an explicit scope (Section 6.6); our failures to establish an in-house task do not show that none exists. Two more specific in-house comparisons were not run: a controlled same-model planted-key comparison, and a team spending weeks on a result before posing that same result under a short budget.

Hindsight. The taxonomy of walls was proposed in c48, after the last design, by the team that had built the failed designs; it was not registered in advance. Several rejections fit it only loosely: the c43 references did not converge (the key itself failed), the c45 Toda chain failed its own early-window assumption, the T1 clue worlds printed the link on the task card, the matrix lane was already beaten by a public file, and the reuse campaign found no reuse advantage. The walls also overlap. Walls 1 and 2 both describe a solver whose repertoire already contains the structure, and walls 3 and 4 both end in a standard method plus speed. A coarser reading fits the record equally well: two walls (repertoire overlap; standard computation plus resources) and one information condition (K3). Other readings are possible.

Self-report.

- The record was written mostly by the AI systems that did the work.
- This paper was drafted by one of them, Claude (directly and through Claude subagents), which also made many of the errors it reports.
- To limit this, every number in the paper carries a note naming the file in the record it was taken from. An independent fact-check of the stage-1 chronicles checked 973 claims: 942 were confirmed, 26 corrected and 5 found unsupported (Appendix F).
- The fact-checkers were themselves AI agents, and they shared the authors' blind spots.
- Pro reviewed the full draft adversarially twice (c49 and c50), and its corrections were applied, among them a weaker statement of the central conjecture and of why the last family was closed. Pro was itself one of the designers, so these were a partner's reviews, not external ones.

Gaps in the record.

- No export exists of the ChatGPT Pro chats or of the Codex operator chat after 8 October. Those rounds are known only through the files they produced.
- Two Claude sessions were deleted.
- Some raw outputs of internal review workflows (eleven workflow records) were not reread.
- Few or no messages survive from some days (12–16 October).
- There is no complete accounting of tokens, cost or calendar effort per round.

No human baseline. We never ran a human arm. Statements about what humans can do rest on the history of science and on outside work, not on our measurements.

Unstable timings. CPU-time measurements on the hybrid PC moved by a factor of 1.41 between sessions for the same work, and by up to 2.3 between core classes. The decisions based on them (Section 6.3) hold under either value, but the individual timings should not be read as precise.

10 Conclusion

We set out to build a benchmark that measures full, autonomous scientific research in physics and that a model could not pass without genuinely improving. Over 29 days (23 September to 21 October 2026) and about 48 rounds, a human lead, who conceived and drove the project, and three AI systems built, ran and then rejected a long series of designs. Read in hindsight, many rejections fall into four overlapping groups by how the answer key was built (the four walls of Section 6), and some had more local causes. The one systematic model failure we found, poor far forecasts without the law family, was not shown to be specific to the model: four generic public-information baselines, tested after the fact, did worse overall.

We do not conclude that such a benchmark is impossible. We conclude that its central difficulty, for a team like ours, was the answer key rather than the task. Producing keys was easy. We never established an in-house family that combined validated scoring, attainability from the supplied evidence, resistance to the declared shortcuts, and useful headroom for the tested model. We state this as an empirical conjecture, with a test that would count against it (Section 6.6). Two specific in-house comparisons stay untested: a controlled same-model planted-key comparison, and a team spending weeks on a result before posing that same result under a short budget. Outside keys, from researchers' unpublished results, from nature or from far larger compute, are worth exploring, but our record establishes neither that outside provenance is necessary nor that it is sufficient, and outside calculation sets already saturate (Section 8.1).

Along the way we learned three rules that we would give any benchmark builder:

1. Programs can reject a task; only model runs can measure whether a remaining gap is closed, and even they do not prove that insight was used.
2. A speed gap is not research headroom.
3. The gap must be measured at the scored precision against an engineered floor.

We also saw the AI systems building the benchmark show failures resembling those the benchmark was meant to measure. They let conclusions run ahead of evidence, lost the goal over long horizons, repeated old ideas as new, and kept procedures out of habit. The retained cases document corrections by the authors' own checks, by other agents' reviews, by mechanical checks and by Erin, and recurrences after written countermeasures. They were selected after detection and have no untreated comparison, so we cannot estimate how many errors any of these prevented. The last review of this paper is an example: Pro's review (c49) found that our first draft of the conjecture, and our account of why the last family was closed, claimed more than the record showed.

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Appendix A Round index

This appendix lists every numbered round of the project, the Codex runs that sat between the rounds, and the Codex task sets, in date order. It is the master index that the project itself never kept.

What a round is. A *continuation* (a *round*; "c17" means continuation 17) was one turn of a hand-over loop. An agent received a package and a state file from the previous turn, did one bounded piece of work, and wrote a new package, which Erin relayed to the next agent. Pro wrote the first numbered packages, which its own chat calls installments. Claude took over at c17 and continued the numbering. From then on Claude (decisions and native checks) and Pro (designs and audits) mostly alternated. Codex, the building and running agent, worked between the rounds. Its runs carry the folder name of the campaign or pilot they belong to, not a round number.

Numbers without a folder.

- **c00.** The record uses c00 for two things. The context bundle sent to Pro files the 23 Sep handoff package as "c00 handoff". A Codex audit of 26 Sep reviews "installments 00, 02–05, and 07"; the only Pro package before c02 is the 25 Sep research dossier, so we read installment 00 as the dossier. The table lists both.
- **c01.** No package, folder or file in the record carries this number. The nearest events are the 25 Sep research reset and the dossier, which the table lists unnumbered or as c00. We do not assign c01 to any event.
- **c06.** Pro's reply was text only because its tools failed, so there is no package. c07 says so, and a later Codex reading confirms it was not a lost file.
- **c13.** Pro's reply was a text-only "pending installment 13" with no package. Codex's validation-only campaign13 filled the slot, and Pro's c14 reviewed that campaign.
- **c31.** Pro's code and tests were lost when its sandbox expired. Only 10 design documents survived, as a snapshot kept at the project root and inside the c33 source records.
- **c35 and c36.** c35 was written and sent to a Pro chat that stalled, and no c35 output is in the record. c36 was never used.

Folder naming.

- Pro, c02–c07: `agi_continuation_NN` in the handoff folder.
- Pro, c08 onward: the unpacked ZIP in a wrapper, `agi_benchmark_continuation_NN/agi_continuation_NN`. From c37 Pro's packages survive as a ZIP plus an extracted copy (c37 inside the c38 check folder).
- Claude, c17–c34: `agi_benchmark_continuation_NN`. A letter suffix (c29a–d, c30a) marks a same-number follow-up, and `c34_post_pilot` holds the work after the T1 pilot.
- Claude's native checks from c38 on: `cNNcheck` at the project root, outside the handoff folder.
- Codex: `work/campaignNN_...` for the quantum pilot. Campaigns 10–13 carry the installment number they served; campaigns 14–18 are resumptions of one pilot. `work/jump_v0X_...` holds the JUMP runs. `codex_tasks_2026_10_20`, `..._20b` and `..._20c` hold task sets 1–3. No campaign19 folder exists; the name appears only as a schema label inside campaign18 files.

Conventions. Dates are the local dates that each round's own document states. Codex reports use UTC, and the table gives UTC only where a time of day matters. The time stamps of Pro's first turns in its chat export are identical, so they are not used. Some Pro documents carry a date one day earlier than the file times of their extracted copies (c20, c22, c28). The Author column names the agent that wrote the package

or ran the work; "+" means a Codex run implemented the round. Erin's relays and go-ahead decisions are mostly not listed; a row mentions them only where its source does. Outcomes repeat each primary source with its own qualifiers; they are descriptive counts, not rates. Model-run counts are in Appendix B.

Terms used in the table. AUTO, HINT and AXIOM are the three help conditions of the JUMP worlds: no help, a neutral note, and the principle stated in words. A *hall* is one of the two experimental facilities (data sources) of a hidden-unity world, and U2 cells are a second variant of its unity questions. OPEN, O0, O1 and L1 are arms of later studies: OPEN gives neither candidate laws nor weights; O0 gives local data only, O1 adds two observed clue values, and L1 also gives the equations and a prior. OVERCLAIM, CLUE_BENEFIT and PARTIAL are labels defined by the reports they come from. A *HOLD* is a halt of a run for an infrastructure problem, settled by a written ruling before the run resumed. N24 is the 24-site case of a quantum ladder of sizes, and `STOP_EARLY_WINDOW` is the frozen rule that stops a design when its cheap-window check fails.

Table 2. Round index, c00 to c48, with the Codex runs and task sets that sat between the rounds.

Round	Date (2026)	Author	What was proposed or run	Outcome
c00 (handoff)	23 Sep	Pro	Rewrote Erin's request for a very hard benchmark into a research handoff and start prompt for Codex.	Package prepared; it started Codex's first study (the fixture study).
Fixture study	23 Sep	Codex	Corpus audit, literature scan and a local pilot of twelve development fixtures.	Ordinary baselines solved all twelve; no valid benchmark established.
Pro review 1	23 Sep	Pro	Independent audit of the fixture study, reproducing its pilot; chose the next experiment.	Reproduced 12/12 baseline passes with no live model; next target was discovery in an existing environment.
DiscoverPhysics study	23 Sep	Codex	Open-discovery experiment over pinned public code; four model workers submitted predictive programs.	All four passed every target; kept as solved calibration.
Pro review 2	23 Sep	Pro	Audited the DiscoverPhysics study, built a field-history prototype, set the next study.	DiscoverPhysics study kept as solved; at most eight new solver attempts allowed.
Field-history study (H1/H2)	24–25 Sep	Codex	Hidden-history cases H1 and H2 plus a clean control; model workers submitted programs.	Both substantive cases solved; kept as development calibrations.
Research reset	25 Sep	Pro	Diagnosed why every design was solved; froze the simulator branch.	Redirected to failure-first search on external tasks.
Headroom study	25 Sep	Codex	Five solver attempts on four external science jobs (Terminal-Bench-Science).	One initial failure, passed on fresh repeat; no repeatable weakness; branch stopped.

Round	Date (2026)	Author	What was proposed or run	Outcome
c00 (research dossier)	25 Sep	Pro	Dossier: revised measurement design, audit of the headroom study, three candidate campaigns.	Specification only; no frontier-hard benchmark demonstrated.
c02	25 Sep	Pro	Built Candidate A (a transfer task); diagnosed the earlier A04 failure against the A05 pass.	A generic client passed 216/216 configurations; kept as negative control.
c03	25 Sep	Pro	Replayed the 25 original assertions; built a crossed-measurement lab; tried a simple A04 repair.	A04 24/25, A05 25/25; repair failed; lab solved 72/72 by the star client.
c04	25 Sep	Pro	Commissioning-service task: repair a wrong evidence program; 162-world grid.	Author reference passed 162/162; passive information bound corrected to 75/162.
c05	25 Sep	Pro	Self-audit after an outside review; new flagship, verified constructive research missions.	A known-family one-burst specialist (author-written) solved the c04 task in 1,000/1,000 worlds; c04 demoted to calibration.
c06	25–26 Sep	Pro	Text-only reply; chose a quantum-memory mission.	No package; target named without running the published reference.
c07	26 Sep	Pro	Implemented the quantum-memory mission on a published code; added a module-placement construction.	Original mission already 186/186; construction span 76 against generic 75.
Pro-handoff resume	about 26 Sep–2 Oct (sources differ)	Codex	Read c02–c07, audited the arc, reproduced the quantum construction, searched for stronger specialists.	Found a cost-74 construction and a generic cost-67 point with no triples; kept as calibration.
c08	2 Oct	Pro	Audited Codex's cost-74 and cost-67 constructions; recovered a matrix-multiplication reference.	Old quantum target closed as solved; no hard item admitted.
Mission admission	2 Oct	Codex + 7 Sol agents	Screened bin-packing, matrix and protein-ranking missions before any solver run.	Nothing admitted; ranking baseline Spearman 0.052 against 0.580 for the negative original row index.
c09	2 Oct	Pro	Audited the mission admission; tested stronger comparators on protein ranking.	Ranking kept as calibration only; packing retired; no flagship.
c10	3 Oct	Pro + Codex	Designed an eight-slot donor and recipient reuse study; first Codex attempt (campaign10).	References reproduced; execution boundary not established; all 8 episodes not started.

Round	Date (2026)	Author	What was proposed or run	Outcome
c11	3 Oct	Pro + Codex	Replacement execution route (offline Docker); first donor episode (campaign11).	Donor ran; a cleanup bug halted the runner; 7 episodes not started.
c12	3 Oct	Pro + Codex	Validated the donor; ran the artifact-only recipient (campaign12).	One episode ended at the budget limit without a commit; the runner classed this as an infrastructure interruption and halted; 6 slots not started.
c13	3 Oct	Codex (Pro text only)	Pro sent no package; Codex ran a validation-only campaign13 to make budget expiry a terminal state.	Correction ready for review; not installed.
c14	3 Oct	Pro	Reviewed campaign13 against 116 supplied tests plus 46 added tests.	Approved with two corrections; activation path supplied.
Campaigns 14–18	3–7 Oct	Codex	Resumed the eight-slot pilot through interruptions and reconstructed contexts.	Seven of 8 slots committed; fresh 2/2, full product 2/2, artifact-only 1/2, donors 2/2.
c15	8 Oct	Pro	Closed the pilot; reviewed a supplied repository; began a second-domain reference.	Closed as COMPLETE_WITH_DISCLOSED_DEVIATIONS; the “fewer than 355 operations” target already solved.
Mission admission 15	8 Oct	Codex + 7 Sol, 1 Luna agents	Offline admission test of the 4x4 matrix-multiplication lane.	Not admitted; several targets already solved; joint cost-and-growth target open.
c16	8 Oct	Pro	Recursive composition of the 48-product kernel to improve coefficient growth.	A constructive reference, not a benchmark item.
c17	8 Oct	Claude	Took over; built JUMP v0.1, disguised revolutions, with a pilot protocol.	Instrument built; pilot handed to Codex.
v0.1 pilot	8 Oct	Codex	Nine blind episodes: revolutions, controls and an axiom comparison.	All 9 passed (96/96 predictions); axiom gap 0; no weakness exposed.
c18	8 Oct	Pro	Adversarial audit of v0.1; positioning check; mathematics for v0.2.	Found a noise-level leak (Claude’s error); v0.1 reframed as a development screen.
c19	9 Oct	Claude	Interpreted v0.1 and the audit; designed v0.2 twin worlds.	Accepted most of Pro’s audit; Codex sent to build and run v0.2.
c20	9 Oct	Pro	Audited v0.1 traces, twin physics and identifiability; critiqued Claude.	Twin physics largely correct; one postulated coupling flagged.
v0.2 pilot	9–10 Oct	Codex	Fourteen-cell ladder of twin worlds, built 9 Oct and scored 10 Oct.	13/14 worlds solved; only miss in the quanta twin, a wrong frequency-law mixture.

Round	Date (2026)	Author	What was proposed or run	Outcome
c21	10 Oct	Claude	Interpreted v0.2 and c20; specified the Hidden Unity probe.	Probe sent to Codex; Pro asked for a Connections design.
c22	10 Oct	Pro	Audited v0.2 and the unity probe; designed a Connections construct.	v0.2 scores reproduced; a first-measurement claim not supported by prior work.
v0.3 hidden unity	10–11 Oct	Codex	Five episodes: can Sol notice a hidden shared law across two halls.	All 5 solved; all 5 U2 cells excluded before any model trial.
c23	11 Oct	Claude	Proposed the Discovery Horizon instrument with a size dial; calibration spec.	Codex sent to run v0.4-cal; Pro to design Horizon v1.
v0.4-cal	11 Oct	Codex	Fifteen episodes: three independent modules at max effort and low effort.	Max effort solved all; low effort solved 4 of 6 modules (inconclusive).
c24	11 Oct	Pro	Coupled-lab Horizon v1 design; critique of the direction; audit of v0.3.	Library-aware reference solver passed 4,080/4,080; coupled pilot proposed, then paused in c25.
c25	11 Oct	Claude	Answered Erin's question whether the project was failing; named the designer ceiling; pivoted to scoring the landing.	Designers can only build what they can check, and Sol had solved it so far.
c26	11 Oct	Pro	Proposed Discover and Land as a measurement layer on research tasks, without adopting "leaps well, lands badly" as a conclusion; audited v0.4.	Adopted as a measurement layer, not a new flagship.
v0.5 landing	11–12 Oct	Codex	Twelve episodes: stated 90 percent ranges when the decisive hall is removed.	Hall A only: 0/18 far truths covered; both halls: 18/18 (three episodes per cell; a small screen). OVERCLAIM label.
c27	12 Oct	Claude + Codex	Interpreted v0.5; specified v0.6 foundations; Codex built them offline.	Foundations ready for Pro's review; nothing launched.
c28	12 Oct	Pro	Audited v0.5 records; vetted coherence elicitation; froze a v0.6 spec.	Admitted, conditional on gates G0–G5 and explicit authorization, a finite-envelope v0.6 pilot of up to 36 episodes.
c29	13 Oct	Claude	Decision and freeze amendment, so that v0.6 bears on the v0.5 failure.	Sent to Pro for a blocking review.
c29a	13 Oct	Claude + Codex	Accepted all of Pro's blocking-review corrections; froze v0.6 and launched Stage 2.	HOLD after 19 contexts: a file exceeded the 16 MB transport cap.

Round	Date (2026)	Author	What was proposed or run	Outcome
c29b	13 Oct	Claude + Codex	Resume decision after the first HOLD; transport repair; run resumed.	HOLD at slot 31: harness error while the coordinator interrupted a turn.
c29c	13 Oct	Claude + Codex	Second resume decision; run resumed.	HOLD at slot 32: Docker image inspection timed out after 60 s.
c29d (v0.6 pilot)	13–14 Oct	Claude + Codex	Third resume; run completed and scored 14 Oct.	36 slots, 38 physical attempts; OPEN regret 0.384, U 0.009, D near 0.
c30	17 Oct	Claude + Codex	Asked whether the OPEN gap is an AI shortfall; Codex ran public-information baselines.	Post hoc, exploratory: Sol beat all four public-information baselines in pooled loss (1.163 against 2.519 for the best), though two simple baselines slightly beat it in Q; this does not establish an AI-specific weakness.
c30a	17 Oct	Claude	Addendum: a reportable v1 "Landing Horizon" with clue worlds and a score G.	Sent to Pro as the c31 prompt.
c31	17–18 Oct	Pro	Audited v0.6; critiqued the score G; designed clue worlds and a small pilot.	Design accepted by Claude; code and tests lost, 10 documents survived.
c32	18 Oct	Claude	Ruled on c31; moved to a new Pro chat; set the c33 question.	Kit issued; Pro turns must end with a ZIP.
c33	18 Oct	Pro (new chat)	Rebuilt the T1 clue pilot with a blind floor, leak probe and Codex package.	Local release verified; native admission pending.
c34	18 Oct	Claude + Codex	Native admission of the rebuilt T1 pilot on Erin's PC; Claude rulings.	Admitted and ready for Erin's go-ahead.
T1 pilot	18 Oct	Codex	Twenty-four frozen episodes: local data (O0), plus clues (O1), plus laws (L1).	Formal reading `CLUE_BENEFIT` (descriptive row `PARTIAL`); mean loss O0 0.665, O1 0.090, L1 near 0.
c34 post-pilot	18 Oct	Claude	Checked the T1 pilot independently; wrote the ambition memo and the Luna check rule.	Everything Codex reported reproduced.
M5	18 Oct	Codex	The same 24 T1 episodes on Luna as a validity check.	Luna worse on O0 and O1 (O1 0.396, worse in 6 of 8 worlds); L1 ties.
c35	18 Oct	Claude, Pro (old chat)	Specified a T3 and T2 matched-arms build for an old Pro chat; the planned Luna screen of c35 material was cancelled the same evening.	The Pro chat stalled and c35 itself was superseded; no c35 output found.
c36	19 Oct	none	Number planned, then skipped.	Never used.

Round	Date (2026)	Author	What was proposed or run	Outcome
c37	18–19 Oct	Pro (new chat)	Discovery Horizon spec and feasibility prototypes; brief by Claude on 18 Oct.	Pro rejected its own pilot: a history learner was exact on all 576 targets.
c38	19 Oct	Claude	Seven-agent native check of c37.	Rejection confirmed: generic learner exact on 576/576; no real dependency; 13 admission gates listed.
c39	19 Oct	Pro	Six families under a kill rule fixed in advance; F1 and F3 prototyped, F2 preflighted, F4–F6 not run.	Nothing admitted: F1 exact on 30/30 target energies across 10 packets (slowest packet 0.034 CPU s on Pro's host); F3 exact 10/10 (under 1 CPU s per packet on Pro's host).
c40	20 Oct	Claude (Fable 5.1)	Replicated c39 natively; diagnosed why the coupled or later discovery candidates (the c24 coupled lab, c37, the c38 repairs D1 and D2, and c39 F1 and F3) never reached a model run.	Rejections stand (manifest 108/108; replay on Erin's PC: F1 30/30, at most 0.172 CPU s; F3 10/10, 0.72–1.31 CPU s); rule "programs reject, models certify".
c41	20 Oct	Pro	Froze a first computational-physics class with a compute asymmetry.	Nothing admitted: a standard method solved the N24 quantum case in 54.7 CPU s on Pro's host (cap 60).
Codex set 1	20 Oct	Codex	Ising floor against exact truth; machine capacity; metered harness; c39 replication.	Floor exact on 90/90 targets (30 packets); no headroom on this ladder.
c42	20 Oct	Claude	Native check of c41 and set 1.	Nothing admitted; N24 took 61.1 CPU s uncapped on Erin's PC (it stopped itself at 60.6 s when capped) against the 60 s cap, 54.7 CPU s on Pro's host: a speed gap.
c43	20 Oct	Pro	Two scale-crossing families, a bilayer and a roughening model, prototyped at reduced scale.	Neither survived; roughening reference radius 0.037 against a 0.01 limit.
Codex set 2	20–21 Oct	Codex	Windows process-tree meter; reach of cluster Monte Carlo and chain integration.	Meter admitted for native work; maintainer reach only extrapolated.
c44	20–21 Oct	Claude (3 reviewers)	Checked c43 and set 2; defined the gap G against an engineered floor.	Nothing admitted; first-pilot bar $G \geq 20$; stop rule proposed.
c45	21 Oct	Pro	Weakly perturbed Toda chain with conserved-charge memory, aiming at $G \geq 20$.	`STOP_EARLY_WINDOW`: memory 0.946869 at t = 16, outside the band [0.975, 1.025].
Codex set 3	21 Oct	Codex (message by Claude)	Message to measure sandbox compile speed, read isolation and one real night's compute.	Never executed; Codex out of quota.

Round	Date (2026)	Author	What was proposed or run	Outcome
c46	21 Oct	Claude (3 reviewers)	Native check of c45; weaker-coupling test; applied the stop rule.	Negative reproduced (own code 0.9475); stop rule fired; options A and B sent to Erin.
c47	21 Oct	Pro	First family of a research-progress instrument: 3D spin-glass ground states.	Exact energy scorer works; variance-only score killed by counterexample; incumbent not admitted.
c48	21 Oct	Claude	Checked c47; saturation probe of the spin-glass family.	Closed by decision (no optimality certificate): at L = 8 and 10 all runs reached the same best energy within 3.44 s of a 30 s budget; at L = 16 runs of one plain parallel-tempering code (no cluster moves) differed by at most 12 units in a bracket about 5,000 units wide (best energy about −7,300 to the trivial bound −12,288), about 0.2%, with best states late in the budget; better algorithms untested.

A.1 Details of selected rounds

This section keeps the detailed accounts of rounds that the main text summarizes, with their source notes.

A.1.1 Phase 1 (c00-c16)

The handoff (c00). On 23 September Erin drafted a prompt asking Codex to build a very hard benchmark for AI that included physics and the human "jump". Pro advised against sending it unchanged, because it risked "an impressive-looking benchmark that measures the wrong thing". Pro wrote a handoff package instead: a research contract, three rival designs and a map of 31 primary sources. Earlier unrelated AI chats of Erin's are not described here. The contract required competent baselines and a report of "all evaluation attempts, budgets, retries, failures and exclusions".

Deterministic baselines passed all twelve of Codex's development fixtures (12/12). Pro's audit found that the science fixtures supplied the equation family and a menu of 256 coefficient possibilities, which removed the discovery step. Fresh gpt-6-sol/high attempts then passed the public DiscoverPhysics extension [38] (4/4 attempts, including a fixed-evidence replay that ran no new experiments). They also passed Codex's field-history worlds H1 and H2: of eleven launches, four stopped at a usage limit and one ended uncommitted, and the six completed artifacts all passed, one with a learned response kernel and no explicit wave equation. Pro showed that the targets were insensitive to the intended dynamics: a predictor with the spatial field frozen still passed the DiscoverPhysics L17 targets 9/9, one with constant accelerations passed L93 6/6, and each of twelve single-trace fits, with the representation supplied, passed all twelve H1 targets. On 25 September Erin asked whether the team was chasing a dead end. Pro admitted it had "improved the machinery faster than we improved the research question".

Headroom study and continuations 02-09. A headroom study of four Terminal-Bench-Science tasks [34] with Astra at high effort showed no repeatable weakness: the one failure passed on a fresh repeat. Pro's installments c02-c09 ran no subject-model episodes. The tasks they proposed fell to generic or specialist programs: a generic client in 216/216 service-transfer configurations, a known-family, one-start specialist in 1,000/1,000 fixed-seed queue worlds, and the natural ordering in 186/186 quantum-memory placement cases on a published [[36,6,6]] code. Codex's ordinary swap-and-anneal search then reached cost 67 in one recorded run, below the earlier archive point of 75. Pro's self-audit said it had "repeatedly selected problems because I could formalize and verify them quickly".

Research-product reuse campaign (c10-c15). Pro's design asked whether a donor's research product helps a later solver, using eight gpt-6.1-sol/max episodes, each capped at 3,600 s and 128 checks. The first attempt failed because a fresh worker could read files outside its workspace; no episode started. Over 3-7

October a Windows cleanup bug, a timeout misfiled as an infrastructure fault, a float error, usage limits and a client update each stopped or altered the run, and four episodes ran under reconstructed budgets. Seven of eight episodes succeeded: donors 2/2, fresh solvers 2/2, full research product 2/2, artifact only 1/2. Both fresh solvers succeeded, so reuse showed no completion advantage in these two blocks. The final report found that the task needed formalization, exact checking and combinatorial search, with no new theory. Pro warned that infrastructure risked "becoming the visible research output".

Matrix multiplication (c15-c16). The published 4x4 target (fewer than 355 auxiliary operations) was already beaten by a current repository file that needed 284, and Codex's offline admission did not admit the lane. Pro's c16 recursive construction has prior art [5] and was filed as a solved constructive reference, not an admitted frontier-hard task.

A.1.2 v0.1 and v0.2 (c17-c22)

v0.1: disguised revolutions. The three families echoed Planck 1900, Einstein 1905 and Einstein 1915. The textbook, curve-fit and Vulcan-patch baselines solved no revolution world. Codex then ran Sol at maximum reasoning effort (the highest setting of how long the model may think): 9/9 episodes, 96/96 predictions and an AXIOM-minus-AUTO gap of 0. Pro's audit (c18) found that the public noise level, printed as round(0.001 C, 5), revealed the hidden speed C, and that an author-written program solved 200/200 kinematics cases with one lab call. Claude's leak test had checked only literal equality. No inspected Sol script used the leak, although absence from the trace is not proof that the public number had no influence. Claude first read the result as recognition ("Sol didn't invent anything"). Pro's trace audit found candidate models generated, fitted and revised, and could not tell recall from derivation, so Claude withdrew that reading in c21. The record supports that Sol solved the v0.1 worlds, not how it did so.

v0.2: twin worlds. Each famous revolution got a non-historical twin on an identical public face. Sol solved 13/14 worlds (159/162 predictions). The miss was the quanta twin (Q-N): Sol derived the energy-ceiling kernel, then averaged two frequency laws whose fits had chi-squared 22.355 and 22.021, and missed three beyond-reach items. Claude concluded the question was unanswerable because the laws agreed within 0.08% inside the lab. Pro's replay showed that this check used temperatures 1 and 20, outside the legal range 1.232 to 15.3; within the legal range, numerical search found a separation of about 0.352%. That refuted Claude's claim but did not show that a cold solver could pick the right law, so Pro advised counting the miss as neither success nor proof of failed invention. The result stays a scored extrapolation miss, after genuine model construction and a consequential averaging choice, under unresolved uncertainty about the structural model; it is neither a success nor a demonstrated identifiability failure.

A.1.3 c37 and c38

The *Discovery Horizon* pilot used exact-rational "conversion cells" with private production rules, matched controls and a proper quantile score, with 24 measurement scalars per stage. Pro's own prototypes killed it. A numeric short-history table predicted all 576 held-out target values within the 24-scalar budget, so the pilot's headline gain over that table could not be positive. Pro recorded `FAIL_PILOT_CEILING`. Claude's seven-agent check confirmed this. An independent learner written only from the public specification was exact on 576 of 576 held-out rows with at most 15 scalars per stage. The ceiling was also predictable by arithmetic: a blind measurement route needed about 12 scalar readings for the new targets, and the independent learner used at most 15 per stage, both below the budget of 24; across 164 cells the scored behavior also collapsed to 20 distinct 12-target vectors. Two exploratory repairs also failed. In D1, targets lay beyond the measurable window, and a generic two-compartment fitter matched them (maximum row loss 1.5e-13). In D2, earlier cells were reused out of regime, and a generic order-2 history learner was

exact on all 576 rows in every variant. Claude counted at least six synthetic designs that had fallen to a cheap or library solver, "a count, not a theorem". (One agent's test, that swapping an earlier law changed the new-target truths in 0 of 3,840 later stages, was not re-run by Claude.)

A.1.4 c39 and c40

The c39 message came from Erin and Claude together. It set requirements R1-R10 for a candidate family, nine *gates* G1-G9 (pass/fail tests a candidate had to clear before any build, condensed from c38's list) and a *kill rule* fixed in advance: a family that failed G1-G6 was not developed further. Pro froze its predictions before writing code. Of six families, three were prototyped or preflighted:

- **F1, a fresh Ising strip** (128 spins with random fields): exact min-sum elimination, a dynamic program that sweeps along the strip keeping one best value per boundary state, returned all 30 target energies exactly, across 10 packets. The slowest took 0.034 CPU s on Pro's host, against a 60 s cap.
- **F2, Silverbox data** (a measured electronic apparatus): dropped because the provider's documentation and scoring code expose the test arrays.
- **F3, graph construction** (degree-3, 24 vertices): a standard mixed-integer linear programming (MILP) solver found 10 of 10 optima in under 1 CPU s per packet on Pro's host (0.709-0.838 CPU s).

F4-F6 (post-cutoff replay, prospective prediction, work samples) were not prototyped. Nothing was admitted, and no subject model ran. Claude's native replay in c40 on Erin's PC matched all 108 manifest entries. F1 was again 30 of 30 exact, with a maximum of 0.172 CPU s, and F3 was 10 of 10 with gap 0 at 0.72-1.31 CPU s. Times differ between the two machines, but the decisions did not.

Appendix B Model-run census

This appendix lists every subject episode that the record shows and summarises the other model use, so that the counts used in the main text (for example "48 of 49" and "zero model episodes") can be read against one denominator. A *subject episode* is a model attempting a benchmark task as the tested solver. A *logical* episode is one scheduled slot in a study plan. A *physical* attempt is one native model context. The two differ when a slot was retried, continued in a reconstructed context, or replaced after an infrastructure failure. All other model use is *non-subject* and is summarised in Table 5. Dates are UTC and come from the primary run files. Model names, efforts and tiers are the requested settings; no serving snapshot was ever authenticated.

B.1 B.1 Subject episodes

The September episodes used gpt-6-sol and Astra (requested name gpt-6-astra), started by the Codex lead as collaboration sub-agents. The October episodes used Sol and Luna in native Codex contexts. Table 3 covers September and Table 4 covers October.

Labels used below: AUTO means no help, HINT a neutral note and AXIOM the principle stated in words; a control is a world whose anomaly has a mundane cause. A hall is one of the experimental facilities (data sources) in a world. The T1 and M5 arms are O0 (local data only), O1 (two observed clue values added) and L1 (equations and prior also given). The v0.6 arms are OPEN (no candidate laws), U (laws without weights) and D (laws and weights).

Table 3. Subject episodes, 23-25 September 2026 (gpt-6-sol and Astra).

Study	Dates	Model, effort, tier, client	Logical / physical	Outcome
DiscoverPhysics study	23 Sep	gpt-6-sol, high; tier and client not recorded	4 / 4	4 of 4 artifacts passed every frozen target (L17 9/9, L93 6/6, control L42 9/9, replay 9/9)

Study	Dates	Model, effort, tier, client	Logical / physical	Outcome
Field-history study (H1 and H2)	24-25 Sep	Astra (max, high) and gpt-6-sol (high, xhigh); tier and client not recorded	6 / 11	6 of 6 logical attempts ended in a passing artifact (H1 12/12 three times, C0 4/4, H2 8/8 twice); 5 launches ended without an artifact
Headroom study, Terminal-Bench-Science tasks	25 Sep	Astra, high; backend identity unverified	5 / 5	First attempts 3 of 4 passed; A04 failed the pooled continuous ranked probability score, CRPS (0.321788 against a bound of 0.216417); fresh repeat A05 passed (0.156348)
September total	23-25 Sep	gpt-6-sol 7 / 9; Astra 8 / 11	15 / 20	14 of 15 logical attempts ended with a passing artifact

Notes on Table 3:

- The DiscoverPhysics study's attempt index lists 87 committed sessions, but only 4 are model attempts. The rest are 11 deterministic baselines and 72 revision-policy stubs.
- The field-history study had 11 launches under a cap that was raised from 8. Four launches stopped at a provider usage limit (slots 2, 4, 6 and 7), and one was left uncommitted (slot 5). The completion workers continued the unfinished scripts of earlier slots (2 to 6 to 9, and 4 to 5 to 10). They are chains, not independent replicates.
- We counted six logical attempts in the field-history study by treating slot 11 (Astra/high) as the replacement of slot 7 (Astra/max), which stopped before committing. By model, the 11 launches were 6 Astra and 5 gpt-6-sol.
- The headroom study had one more launch of A01, rejected by the agent-thread limit before any exposure. It is not counted. All five attempts shared a host with the grader, so the study was cooperative development, not a blind evaluation.
- The Codex client version and the service tier were not recorded for any September run.

Table 4. Subject episodes, 3-18 October 2026 (Sol and Luna, native Codex contexts).

Study (round)	Dates	Model, effort, tier, client	Logical / physical	Outcome
Research-product reuse campaign (c10-c15)	3-7 Oct	Sol, max, default; codex-cli 0.160.0, then 0.162.0-alpha.2	8 / 13	7 of 8 committed (donor 2/2, fresh 2/2, research product 2/2, artifact-only 1/2)
v0.1 disguised revolutions (c17)	8 Oct	Sol, max, default; codex-cli 0.162.0-alpha.2	9 / 9	9 of 9 worlds solved; 96 of 96 predictions
v0.2 twin worlds (c19)	10 Oct	Sol, max, default; codex-cli 0.162.0-alpha.17.2	14 / 15	13 of 14 worlds solved; 159 of 162 items; the miss is Q-N
v0.3 hidden unity (c21)	11 Oct	Sol, max, default; alpha.17.2	5 / 5	5 of 5 solved; 115 of 115 items; 5 other cells never launched
v0.4 calibration (c23)	11 Oct	Sol, max (9 episodes) and low (6), default; alpha.17.2	15 / 15	13 of 15 episodes; 25 of 27 modules (max 21/21, low 4/6); 400 of 414 items
v0.5 landing (c25)	11-12 Oct	Sol, max, default (12 contexts) and priority (1); alpha.17.2	12 / 13	Far unity points passed 18/18 with both halls and 0/18 with Hall A only; ranges covered 18/18 and 0/18
v0.6 reporting pilot (c28-c29d)	13-14 Oct	Sol, max, default; alpha.17.2	36 / 38	Mean conditional expected regret 0.384 (OPEN), 0.0087 (U), about 5e-14 (D)
T1 clue pilot (c33-c34)	18 Oct	Sol, max, default; alpha.17.2	24 / 24	Mean loss O0 0.665, O1 0.090, L1 about 0; formal reading CLUE_BENEFIT
M5 Luna check (c34)	18 Oct	Luna, max, priority (Fast); alpha.17.2	24 / 24	Mean loss O1 0.396, O0 3.848; formal reading INCONCLUSIVE

Study (round)	Dates	Model, effort, tier, client	Logical / physical	Outcome
October total	3-18 Oct	Sol 123 / 132; Luna 24 / 24	147 / 156	Last subject episode ended 18 Oct 19:02

Notes on Table 4:

- *Campaign.* The eight logical slots needed 5 additional requested continuations, so 13 task-bearing contexts ran. One earlier empty setup context and one routing canary are counted separately. Slots 3 to 6 were finished in reconstructed contexts built from saved visible histories, each with its unused budget (for example slot 3: 145 + 1,713 + 1,742 s). The client build changed during slot 6.
- *v0.2.* E07 failed on a provider usage limit and was run again in a fresh context under a disclosed retry amendment. That gives 14 logical episodes and 15 physical attempts. Episode E10 used two turns of one context.
- *v0.4.* Two U seeds were replaced by pre-drawn reserves before any learner ran. No scientific episode was retried.
- *v0.5.* Episode E05 lost its first context to a provider-capacity failure after 40 accepted calls. A reconstructed second context made no new lab calls. E05 stays in the primary denominator as one logical episode.
- *v0.5 tier.* The thread record of episode E03 shows the priority tier, and the other 12 contexts show the default tier. The final report states that model and effort were unchanged but does not mention the tier. E03 passed all its points.
- *v0.6.* The 38 physical attempts are 36 slots plus two authorised re-attempts. Original slot 05 returned no readings when Docker was paused during research. Original slot 32 failed at Docker image inspection before any native turn. The replacements 05-R1 and 32-R1 enter the primary analysis, and the late replacement weakens the exchangeability assumption behind the reported p-value.
- *T1 and M5.* Each had 8 worlds in 3 arms (O0, O1, L1), with no replacements. M5 was first held at admission, with zero inference, because the effective service tier (priority) differed from the one used in T1 (default). It then ran on priority as a declared deviation.
- *Why 176 exceeds 162.* The 14 extra physical attempts are the 5 lost field-history launches, the 5 campaign continuations, the v0.2 retry, the v0.5 reconstructed context and the 2 v0.6 re-attempts. The empty campaign setup context is not included.
- *Resources.* No dollar cost is recorded for any run. Token counters exist only for some studies, for example 2,319,906 native tokens over the nine v0.1 episodes and 5,626,288 over the 13 v0.5 contexts.

B.2 B.2 Non-subject model use

Table 5. Non-subject model use. Counts are lower bounds.

Use	Dates	Model and effort	Count	Notes
Codex authoring and review (first September stage) and the Pro hand-off resume	23 Sep; 26-27 Sep	Lead Astra max; workers gpt-6-sol high, then Luna max and gpt-6-sol xhigh	3 workers; then 1 Luna and 2 Sol	Corpus extraction, reproduction, audits. No solver ran.
Codex mission-admission waves	2 Oct; 8 Oct	Sol max; one Luna max on 8 Oct	7 agents; 8 agents	One extra spawn on 2 Oct was rejected. No capability episodes.

Use	Dates	Model and effort	Count	Notes
Isolation probes and routing canaries	3-18 Oct	Sol max (one Luna context in M5)	1 boundary probe; about 12 routing or status turns	The campaign 10 probe read excluded files. Turns: campaign 1 + 1, v0.1 1, v0.2 4 (two failed), v0.3 1, v0.4 3 (one failed), v0.5 1. The c34 admission used one empty context and the M5 admissions two (the first was held on a tier mismatch, the second was authorised and passed), all with zero turns.
T1 step coder (A6), exploratory	18 Oct	Sol max	1 context, 1 turn	Coded visible records only. No effect on scores.
Codex task sets 1 and 2	20 Oct	Sol max; Luna max	Set 1: 8 agents (5 Sol, 3 Luna); set 2 not itemised	Code development and review. Zero subject episodes.
Pro	23 Sep-21 Oct	Not recorded	Not countable	Designer. Its own sandbox scripts and internal sub-agents (3 in c41) are not model episodes.
Claude workflow agents	18-21 Oct	claude-opus-5-5 (140), claude-haiku-5-5 (1), placeholder (2)	143 transcripts, 11 workflows	Reviews, refutation, c38 checks, stage-1 record reading (33). Claude's own turns are not counted.

B.3 B.3 Totals and reconciliation

Table 6. Subject episodes by month and model (Table 3 and Table 4 summed).

Month	Model	Studies	Logical	Physical
September	gpt-6-sol	DiscoverPhysics study, field-history study	7	9
September	Astra	Field-history study, headroom study	8	11
October	Sol	Campaign, v0.1-v0.6, T1	123	132
October	Luna	M5	24	24
Total			162	176

The record therefore holds 162 logical subject episodes and 176 physical attempts. Only 20 physical attempts used the September models, so any comparison of September and October results compares different models. The 8-episode campaign of 3-7 October was not the first model campaign, because 15 logical episodes had run in September.

Designs proposed from c37 on. Zero subject episodes ran on any design proposed from 19 October (c37) on. The designs concerned are c37 (brief 18 Oct, delivered 19 Oct) to c48, Codex task sets 1 and 2, and the never-executed set 3. The c37 pilot was rejected by its own test before any build, and the reports of c39 to c46 state that no subject model ran. For c47 the contract was no subject runs, and c48 used only Claude's own optimiser. Codex set 3 was never sent. The last subject episode ended on 18 October at 19:02, and no later native transcript exists outside packaging copies.

What "48 of 49" counts. The tally is v0.1 9/9, v0.2 13/14, v0.3 5/5 and v0.4 21/21. It counts 49 scored worlds or modules from 37 episodes at max effort. A v0.4 institute episode holds three modules, so the 9 max-effort v0.4 episodes (3 single-module and 6 institute episodes) give 21 module attempts. It leaves out the 5 unlaunched v0.3 cells and the six low-effort v0.4 modules (4 of 6 passed). With those six modules added, the same arithmetic gives 52 of 55.

Table 7. Cells behind "48 of 49".

Study	Cells	Passed	AXIOM, HINT or control cells	Notes
v0.1	9	9	6	3 controls, 3 AXIOM
v0.2	14	13	6	3 controls, 3 AXIOM; Q-N missed

Study	Cells	Passed	AXIOM, HINT or control cells	Notes
v0.3	5	5	2	1 HINT, 1 AXIOM-U
v0.4, max effort	21	21	9	9 AXIOM modules
Total	49	48	23	26 cells are plain AUTO

Our cell-by-cell count gives 23 assisted or control cells, and 26 plain AUTO cells without a control role. The project files say "at least 20", which this count satisfies. The c25 decision text states 48 of 49 ("at max") without the assisted-cell qualifier. The figure also mixes worlds with modules, reuses physical seeds (v0.3 presented two physical worlds in five contexts, and v0.4 matched seeds across conditions) and is descriptive, not a rate.

What "zero model episodes" means. ROAD_AFTER_C39 says that in its "21 rounds" (c17-c39) "zero model episodes" ran on any discovery design. It then names the model data that does exist: the early single worlds, the landing tests, the T1 clue pilot, and the campaign before c17. Read literally as "no model ran in c17-c39", the sentence does not hold, because 139 logical episodes (143 physical) ran in v0.1 to M5. The correct reading is narrower. No model ran on the coupled or later discovery candidates (the c24 coupled lab, whose Sol pilot was paused, the c37 pilot, the c38 repairs D1 and D2, and c39 F1 and F3) or on any design proposed from c37 (19 October) on. The landing studies v0.5 and v0.6, the T1 clue pilot and M5 did run.

The same section's table lists the c33 T1 curve worlds among the designs that died, yet T1 and M5 produced 48 logical episodes. Those episodes are calibration data on a building block, so the paper uses the census numbers and not the ROAD sentence.

B.4 B.4 Landing and clue studies in detail

This section keeps the detailed account of the landing and clue studies summarized in Section 4.3 and Section 5.

The v0.5 landing screen. Codex ran 12 Sol episodes in a simulated world with two laboratory "halls". With Hall A only (the hall with the limited frequency range) and no audit instruction, Sol's 90% ranges for targets at 8, 12 and 16 times Hall A's frequency limit covered 0 of 18 truths (all 30 misses in the Hall-A-only cells lay below the lower bound). With both halls, the same model on the same seeds (a seed fixes one random instance of the world) covered 18 of 18. Pro corrected Claude's reading that "the numbers priced no model risk": all six range writers had included some allowance. Pro also noted that Hall A gave 40 scalars where both halls gave 100, so data acquisition may have contributed.

The v0.6 study. Pro designed 12 worlds times three arms; Claude replaced one arm with OPEN, which gives neither candidate laws nor weights. The run used 36 logical slots and 38 physical attempts. Three infrastructure HOLDs (stops called when the run machinery failed) halted it (a transport cap, an interrupt race, a Docker timeout), each settled by rulings written without sight of the scores.

Terms used in the table. The v0.6 arms differed in what the forecaster was told: OPEN gave neither laws nor weights, UNDECLARED gave the candidate laws without weights, and DECLARED gave laws and weights. Delta is expected regret against a reference that knows the candidate envelope (the set of laws the world could follow). Loss is mean proper forecast loss, a score that is best when the forecaster reports its true belief; lower is better. The T1 arms (c33, described below) were O0 (local data only), O1 (two aggregate clue readings, with the clue's identity printed on the card) and L1 (laws given). The Fast tier is a service tier other than the default one.

Table 8. Key landing and clue results. Delta is expected regret; loss is mean proper forecast loss (see the text).

Study	Condition	Result	Note

Study	Condition	Result	Note
v0.5	Hall A only	0/18 covered	3 seeds, no audit
v0.5	Both halls	18/18 covered	same seeds
v0.6	OPEN, Delta	0.384	12 worlds
v0.6	UNDECLARED, Delta	0.0087	laws, no weights
v0.6	DECLARED, Delta	about 5e-14	laws and weights
T1 Sol	O0 / O1 / L1 loss	0.665 / 0.090 / about 1e-16	8 worlds
T1 Luna	O0 / O1 / L1 loss	3.848 / 0.396 / about 2e-16	Fast tier

In OPEN, Delta rose from 0.0015 at 1.2 times the accessible limit to 1.15 at 16 times, and coverage (the fraction of truths inside the stated 90% range) fell from 0.96 to 0.71. The pattern was erratic: one slot covered 10/10 with log-widths (the natural log of upper bound over lower bound) near 106, another covered 4/10. The cause stayed unresolved.

c30: an AI weakness? Claude noted that the reference knew the envelope. Codex built four public-information forecasters and compared them post hoc on the same twelve journals. Sol's pooled loss was 1.163 against 2.519 for the best of them, and Sol beat every one in all six U1 worlds. In Q, all OPEN methods ended between 1.84 and 1.90, Sol included. The report says this "does not establish an AI-specific weakness". It also showed that the c28 controls Claude had cited for OPEN behaviour used the hidden law family.

Clue worlds (c31-c34). Claude proposed worlds where a public clue implies that the law must bend. Pro showed Claude's gap-closed score was not bounded, then specified 8 worlds times the three arms O0, O1 and L1. Pro's code and 81 tests were lost when its sandbox expired. Claude logged six of its own errors. In the rebuild, the "blind" floor author could read the L1 cards, so the floor equalled the reference and no vocabulary-blind floor existed.

Sol then ran 24 of 24 episodes. The formal reading was CLUE_BENEFIT (the clue helped) with exact p = 0.0078125 (2/256, the minimum with 8 worlds). A generic Gaussian-process rung (a standard curve-fitting method with no design knowledge) scored 0.188 on Sol's O1 journals and the design-informed shape library 0.014. Because the card named the clue, T1 tested clue use, not discovery. In two worlds the reference gained nothing from the clue, yet Sol gained 2.386 and 0.204, so part of the benefit was weak use of in-window data.

M5 (Luna). Luna ran the same episodes on the Fast tier while Sol had used the default tier, so the comparison is descriptive. Luna's O1 loss was higher than Sol's in 6 of 8 worlds, and higher than the generic rung on its own journals (0.396 against 0.186; the same rung scored 0.188 on Sol's journals). L1 was exact for both.

Appendix C Arithmetic of the compute gap

This appendix states the cap of Section 6.3 with its assumptions. The definitions are those adopted in c44, after the team had twice mistaken a ratio of sizes or speeds for research headroom.

Definitions.

- The agent must report a target quantity to within a tolerance τ and has a CPU budget B_a. The maintainer, who produces the key, has a budget B_m.
- Let $C(\varepsilon)$ be the CPU cost for the cheapest member of a frozen *standard menu* to reach precision ε. The menu contains:

- an engineered direct method: native vectorized code in a standard layout, pinned to the subject's core class, timed over at least three repeats, with an independent attempt to speed it up;
- standard variance reduction;
- the frozen standard extrapolation;
- the standard effective theory.

- $C_a = C(\tau)$ is the cheapest cost over the declared standard menu, and the *gap* $G = C_a/B_a$ is its ratio to the agent budget. A separate quantity G_{direct} denotes the corresponding ratio for the engineered direct route alone. A lower bound on G concerns every implemented member of the menu; timing only the direct route establishes G_{direct}, not G. Neither shows that research or insight is required.

The cap. The maintainer computes the key with the menu to a radius $r < \tau$, so that the key is precise enough to score against. Then $C(r) \le B_m$, and in general $G = C(\tau)/B_a \le (B_m/B_a)\, C(\tau)/C(r) \le B_m/B_a$. If one method is cheapest over the whole range $[r, \tau]$, with $C(\varepsilon) = c\,\varepsilon^{-\kappa}$, then $c \le B_m r^{\kappa}$ and

$$G = \frac{c\,\tau^{-\kappa}}{B_a} \le \frac{B_m}{B_a}\left(\frac{r}{\tau}\right)^{\kappa}.$$

If the cheapest member changes between τ and r, for example from sampling at τ to an exact method of fixed cost at r, only the general form holds.

The bound assumes the maintainer has no better method than the standard menu. If the better method is published, a solver can recall it, and it belongs in the menu. If it is the maintainer's own, the benchmark tests whether the agent finds that method, which is the designer-ceiling question again (Section 6.6). Either way, the designers choose the menu, so K2 is only as strong as their knowledge of standard methods.

Values of κ.

- **Sampling error:** $\kappa = 2$.
- **Power-law bias:** a bias $AS^{-\omega}$ at scale S, with cost proportional to S^D, gives $\kappa = D/\omega$. Standard extrapolation raises ω and so lowers κ.
- **Exponentially vanishing bias:** a bias Ae^{-aS} with polynomial cost $c\,S^D$ gives a precision cost proportional to $[\log(A/\varepsilon)]^D$, so the effective exponent is $D/\log(A/\varepsilon)$, which becomes small when $\log(A/\varepsilon)$ is large compared with D, provided fluctuations are already resolved at the required precision. If the cost grows exponentially with S, the precision cost can again be a power law. In the favourable case the cap approaches B_m/B_a. Examples are effects outside a light cone, or gapped finite-size corrections. c44 states a regime check for this case, valid under those assumptions, $(\ell_{\text{bias}}/\ell_{\text{fluct}})^d \ge n_m$, where ℓ_{fluct} is the correlation length of the observable's fluctuations and n_m is the number of independent fluctuation units the key needs. In words, one minimal unbiased system must already contain all the independent samples the key requires.

Ways around the cap, and what we tried. Within one machine, the ways around the cap are regimes where the key's cost hardly grows with precision. One is a systematic error that falls off exponentially beyond a scale the agent cannot reach. If the bias is Ae^{-aS} at scale S and the cost grows as a power of S, the precision cost grows only like a power of $\log(1/\varepsilon)$, so the effective κ becomes small when that logarithm is large compared with the cost exponent; this is "scale-crossing", in which the maintainer's larger computation sees physics the agent's cannot. Others are an exact method affordable to the maintainer but not the agent, and a key known by construction (an exactly solvable point or a planted solution), which carries a recall risk. Earlier rounds tried exact finite-system truths (c39 F1, the c41 quantum and Ising targets), and standard methods reached, or nearly reached, the scored targets within the agent's budget. We built no demonstrated bias-limited candidate. The last candidate, the memory of a conserved charge in

a weakly perturbed Toda chain, was registered as variance-limited ($\kappa = 2$) and failed its own frozen early-window check before a native G was measured. Our independent re-check then showed that the assumed scaling with g^2t did not hold at the couplings tested, and that at the reduced packet's own target the memory was not resolved away from zero (0.009 ± 0.087, one standard error; Appendix D).

Table 9. Caps on G under one nominal night of 230,400 CPU s (8 equal cores for 8 hours), from c43's design arithmetic as checked in c44.

Setting	Cap on G
$B_a = 600$ s, $r = \tau/4$, $\kappa = 2$	24
$B_a = 1{,}200$ s, $r = \tau/4$	12
Six instances share the night, $r = \tau/4$, $B_a = 600$ s	4
One reference coordinate at 99% coverage against the floor's 95%, $r = \tau/4$, $B_a = 600$ s	13.9
$B_a = 600$ s, $r = \tau/2$	96
Illustrative joint pass probability ($a^2 = \tau^2 - r^2$; independent unbiased Gaussian errors, r the actual reference noise), $r = \tau/4$ and $\tau/2$	25.6 and 128
$\kappa \approx 0$, $B_a = 600$ s	about 384

The real machine. The PC has 4 performance and 4 efficiency cores, and the same work cost 1.3–2.3 times more CPU seconds on an efficiency core. A night is therefore about 5.7–7 performance-core equivalents instead of 8 before other load, which lowers every cap in Table 9 in proportion (about 17–21 for the first row). CPU times for identical work also moved by a factor of 1.41 between sessions. For that reason no decision in c44 rested on a margin under 2×.

Consequence. Our first-pilot bar was a lower bound $G \geq 20$, against an engineered floor, at a frozen $B_a = 600$ s. At $r = \tau/4$ the hybrid-core cap of about 17–21 left little or no margin over that bar; a looser reference precision raises the cap, but no formula here gives a lower bound on G for an actual candidate. The pass-probability row assumes independent unbiased Gaussian errors with matched confidence multipliers; scoring against a frozen reference, unequal coverage, bias or correlated errors need a different calculation. For variance-limited sampling at the quarter-radius convention, the displayed one-night caps lie below the release aim of $G \geq 100$. Looser reference precision or other pass-probability conventions raise that upper envelope (the half-radius joint-probability row gives 128 nominally, and $128 \times 7.053/8 \approx 112.8$ at the higher effective-core estimate), but none establishes an attainable lower bound for a candidate. We established no candidate at the release aim under the attempted resource and reference rules, and built no bias-limited candidate: every sampled c43 target was variance-limited, and the c45 Toda chain memory, registered with $\kappa = 2$, failed its early-window rule (Appendix D).

C.1 How the accounting developed (c41-c44)

This section keeps the detailed chronology of the compute-asymmetry rounds that Section 4.5 summarizes.

c41 and c42. Pro tested a fermion-chain quench, long-time dynamics of a nonlinear chain (the Fermi-Pasta-Ulam-Tsingou, or FPUT, chain) and disordered transport. An agent attack had 60 CPU seconds; the maintainer had 1,800 per packet. No class was admitted. Standard solvers solved transport, and long FPUT references did not converge. A float32 Chebyshev method (a single-precision polynomial time-evolution method) that computed only the scored outputs killed the 24-site quantum spin-chain case (N24) in 54.70 CPU s on Pro's host, with error 2.26e-8 against a tolerance of 0.001. In Codex set 1, a blind annealing and parallel-tempering floor with 60 CPU s per packet matched the exact ground-state energy on 90/90 targets (30 packets) of square $\pm J$ Ising instances up to 28x28.

Claude replayed N24 natively. The worker used 61.109375 CPU s uncapped (it stopped itself at 60.609375 s when capped) against the 60 s cap. Claude judged this a hardware margin, not headroom: floor and truth were the same computation at the same scale, so the only possible gap was a budget ratio of at most 30, which faster code closes. This became the lesson *a speed gap is not research headroom*. Claude also found that Pro's CPU meter charged 0.015625 CPU s for that worker, because it timed only a launcher process. The next round therefore asked for *scale-crossing* targets, at least about 8 times beyond the agent's reach in linear size, with a value set by physics visible only at that scale.

c43 and c44. Pro's two reduced prototypes failed. The Ising bilayer reference did not equilibrate, a protocol failure Pro owned. The roughening reference was too noisy (radius 0.0370518141 against a limit of 0.01). Claude admitted the meter that Codex set 2 delivered, a Job-Object meter (built on a Windows Job Object) that charges the whole process tree. It charged 43.609375 CPU s for the N24 worker that measured 43.453125 s itself. The same work took 43.45 and 45.09 CPU s in later sessions, against 61.11 s in c42, a factor of 1.41. The PC was also hybrid, with four performance and four efficiency cores. Claude also caught its own error (weakness W13): its equal-error reach specification ignored the answer key's precision premium. For every sampled target in c43 and Codex set 2, the scale gap was a precision gap.

c44 therefore defined the gap as $G = C_a/B_a$, the smallest native CPU time at which any member of a frozen standard menu reaches the scored precision, divided by the agent budget B_a. With sampling error it is capped at $G \le (B_m/B_a)(r/\tau)^{\kappa}$ with $\kappa = 2$, where B_m is the maintainer budget, r the radius of the key's uncertainty and τ the scoring tolerance (Section 6.3, Appendix C). The cap is 24 at $r = \tau/4$ and 600 s on a nominal night of 8 equal cores, and about 17 to 21 once the hybrid CPU is counted. The first-pilot bar was a lower bound $G \ge 20$ plus the (f) facts at a frozen 600 s, with a stop rule, and Erin confirmed it by sending c45. A reviewer objected that the bar of 20 sat inside the range of caps (9.3 to 42.7, depending on convention) and that it readmitted variance-limited targets. On the hybrid CPU the bar was at or above the cap for such targets at $r = \tau/4$. The final report kept them, noting that a win by a novel estimator measures only a piece of research.

Appendix D Two worked cases

These cases illustrate the compute wall (Section 6.3) and the cheap-to-check wall (Section 6.4). Both closed on 21 October 2026, before any subject-model episode ran on either design. "CPU s" means seconds of processor time, and "native" means run directly on Erin's PC rather than in the sender's sandbox. Appendix C defines the gap G used below.

D.1 Memory of a weakly perturbed Toda chain (c45, c46)

Setting. Pro proposed a periodic chain of $N = 64$ bonds with stretches r_i, momenta p_i and

$$H_g = \sum_i \left[\tfrac{1}{2}p_i^2 + e^{-r_i} + \tfrac{g}{4}r_i^4\right], \qquad \dot{r}_i = p_{i+1} - p_i, \qquad \dot{p}_i = V_g'(r_i) - V_g'(r_{i-1}),$$

where $V_g(r) = e^{-r} + gr^4/4$. At $g = 0$ this is the integrable Toda lattice, a chain with exponential nearest-neighbour forces that has extra conserved quantities, among them a third charge Q. States came from the stationary ensemble $\propto \exp[-(H_g + P\sum_i r_i)/T]$ with $T = 0.5$, $P = 1$. With $y_i = e^{-r_i}$, $m = E[y]$ and $v = \mathrm{Var}(y)$,

$$Q = \sum_i \left[\tfrac{1}{3}(p_i^3 - 3Tp_i) + p_i(y_i + y_{i-1} - 2m)\right], \qquad D = E[Q^2] = N\left(\tfrac{2T^3}{3} + 2Tv\right).$$

The *memory* is $C_g(t) = E[Q(t)Q(0)]/D = 1 - E[(Q(t) - Q(0))^2]/(2D)$, which equals 1 at all times when $g = 0$. The task was to forecast C at a remote time to within $\tau = 0.10$, a target that, by design, a night of computing might resolve and the agent's 600 CPU s could not (never established: the design's cost figures were upper envelopes, not lower bounds on G).

Frozen rules. Before scientific implementation, Pro registered that at $t = 16, 32, 64, 128$ every simultaneous 99% interval must intersect $[0.975, 1.025]$, so the chain looks conserved in the cheap window. The reduced packet used $g = 0.04$ and a remote $t = 4{,}096$. The unexecuted native ladder held $g^2 t = 6.5536$ fixed, with g down to 0.00125 and t up to 4,194,304. Pro labelled this scaling a hypothesis to be tested.

The window failed. With 256 paths, $C = 0.9469, 0.9068, 0.8674, 0.7455$ at the four times, and the first interval was $[0.9295, 0.9642]$. All four missed the band. Claude's native replay used 5.9 CPU s. Independent code gave $C(16) = 0.9475$ with interval $[0.9382, 0.9568]$ from 1,024 paths; its $g = 0$ control gave $1 - 2 \times 10^{-9}$.

The scaling failed. Claude then tested whether C depends on $g^2 t$ alone, for $g = 0.01$–0.08.

Table 10. Memory C ($\pm$ one standard error) at fixed $g^2 t$; 256 paths per point, except 1,024 for $g = 0.04$ at 0.2048. $t_{1/2}$ is a rough half-life. a is the local exponent of $t_{1/2} \propto g^{-a}$ between g and $2g$.

g	$g^2 t = 0.2048$	0.8192	1.6384	$t_{1/2}$	a
0.08	0.855 ± 0.015	0.648 ± 0.031	0.493 ± 0.045	248	
0.04	0.748 ± 0.013	0.469 ± 0.051	0.262 ± 0.067	431	0.80
0.02	0.609 ± 0.048	0.342 ± 0.066	0.049 ± 0.091	901	1.06
0.01	0.575 ± 0.043	0.194 ± 0.081	0.205 ± 0.072	2,686	1.58

At fixed $g^2 t$ the memory fell as g fell, so the rescaling did not hold. The exponent rose toward the golden-rule value 2 (the weak-coupling perturbative scaling, in which the decay rate grows as g^2), so a crossover at weaker coupling is plausible but was not shown. If $a = 2$ held below $g = 0.01$, the native targets would lie about 24 half-lives out. At $g \geq 0.02$, points 4–18 half-lives out were consistent with zero. At $g = 0.01$, C was 0.19 and 0.21 at 3 and 6 half-lives, 2.4 and 2.8 standard errors above zero, so a slow tail is not excluded. Weak coupling was also cheap to measure directly (an estimated 0.15 CPU s per path at $g = 0.01$, $t = 16{,}384$), so an agent could extrapolate. The reduced packet's own target was not resolved from zero: $C = 0.009 \pm 0.087$ (one standard error) at $g = 0.04$ and $t = 4{,}096$.

Consequence. Erin's stop rule said that if c45 and its native gate gave no candidate with a lower bound $G \geq 20$ (plus further required facts), the team would stop designing on this route. The gate ran and failed, so the rule applied; no answer key was built. Codex set 3, which would have measured the native machine further, never ran because Codex was out of quota. The result covers one family, one ensemble and $g = 0.01$–0.08.

D.2 Ground states of 3D spin glasses (c47, c48)

Setting. Erin then chose a research-progress instrument. Pro's family was the three-dimensional Edwards-Anderson spin glass. A spin glass is a lattice of spins $s_i = \pm 1$ with random bond signs, so that not every bond can be satisfied at once; its *ground state* is the spin configuration of lowest energy. Here the lattice was a periodic cube of side L (so L^3 spins), with one bond per site in each positive direction. Magnitudes were 1 (isotropic) or, on z-bonds, 2 (anisotropic).

$$E(s) = -\sum_{(i,j)} J_{ij} s_i s_j \;\geq\; E_0 = -\sum |J_{ij}|, \qquad u = \frac{E_{\text{inc}} - E_{\text{agent}}}{E_{\text{inc}} - E_0}.$$

E_0 is the trivial bound, reached only if every bond is satisfied. The incumbent, with energy E_{inc}, is the standard reference method the agent has to beat. The utility u is the fraction of the incumbent's certified interval removed, not of its true error: a frustrated triangle has minimum -1 but $E_0 = -3$.

No answer key. The agent submits a spin vector and the scorer recomputes its energy exactly, so every submission is a valid upper bound. On exactly solved controls, verifying a 4,096-spin packet took a median 0.024 CPU s. The ground state is never needed. The score certifies the state, not optimality.

The variance score. Claude's proposal scored sampling methods by variance. Pro rejected it with exact counterexamples; in one, reproduced natively, 50 independent dimers gave a variance gain of 100 while the mean squared error was $181/1250$ against $1/25$, so a biased method looked efficient and was worse.

Native check. Claude's c48 check found 595 of 595 manifest files correct and re-scored all 8 exposed packets with its own code; every recorded energy matched. At 1 CPU second, Claude's plain parallel-tempering code (below) matched PeaPods PT on all 8 exposed packets. The PeaPods incumbent itself was not replayed; that needed a download requiring Erin's approval.

Saturation probe. Claude wrote a deliberately plain parallel-tempering code (copies of the system run at several temperatures, with neighbouring copies swapping states): 24 temperatures spaced geometrically over 0.2–2.0, single-spin Metropolis sweeps (random single flips accepted by an energy rule) with lookup tables, replica exchange between neighbours, no cluster moves, and best-state tracking. Each run had 30 CPU s, the planned evaluation budget, on one pinned performance core. Packets came from Pro's generator with fresh seeds: four sizes, two regimes, two draws and four runs each (64 runs). Pro's scorer confirmed every energy.

Table 11. Saturation probe, 8 runs per row. "Agree" counts runs reaching their packet's lowest energy. Time is CPU s to a run's own best. Spread is the largest gap among a packet's 4 runs, per draw. Lowest E is for the second draw.

L	Regime	Agree	Time (s)	Spread	Lowest E (E_0)
8	isotropic	8/8	0.00–0.02	0, 0	−922 (−1536)
8	anisotropic	8/8	0.02–0.33	0, 0	−1288 (−2048)
10	isotropic	8/8	0.08–0.34	0, 0	−1792 (−3000)
10	anisotropic	8/8	0.20–3.44	0, 0	−2522 (−4000)
12	isotropic	8/8	1.6–25.1	0, 0	−3084 (−5184)
12	anisotropic	4/8	5.3–26.8	4, 4	−4358 (−6912)
16	isotropic	2/8	18.2–29.1	8, 12	−7312 (−12288)
16	anisotropic	7/8	19.5–26.5	0, 4	−10272 (−16384)

Reading. At $L \leq 10$ all runs agreed within 3.44 s of a 30 s budget. If those energies are ground states, no method could score above zero. At $L = 16$ the isotropic runs differed by at most 12 units in a scored bracket about 5,000 units wide (from the best energy found, about −7,300, to the trivial bound, −12,288), about 0.2%. Their best states appeared late in the budget (18.2–29.1 s), so this plain method was still improving; the spread of one method's runs does not bound what another method could gain. Part of the smallness comes from the normalisation: the trivial bound lies far below the best states found. We did not test whether a better algorithm, rather than faster code, would open more room (for example cluster moves or population annealing); that would be methods research, the thing the research-progress instrument aimed to measure. c47's registered falsifier needed an independent optimality proof, which this probe did not give; the family was closed by decision.

The probe informed the decision not to run the pilot. It did not establish c47's registered saturation falsifier, and engineering or search gains were allowed by c47's narrowed scoring contract. Agreement of four runs is strong evidence, not proof, and the probe used one plain method on two packets per cell. Standard Monte Carlo ground-state methods (population annealing, simulated annealing and parallel tempering) are compared on a three-dimensional spin glass with Gaussian disorder in [35].

Appendix E The AI team's failure log

This appendix reproduces the failure log that Claude kept about itself and adds notable failures of Pro and Codex. The entries were written by the AI that made the errors, so they are self-reports and may be incomplete. Claude created the log on 18 Oct with rows W1 to W10, added W11 that evening, added W12 on 20 Oct and W13 later that day, and kept adding instances until 21 Oct.

The log was therefore written during the project. Instances dated before 18 Oct, such as the c17 warning under W1 and the c21 and c30 claims under W3, were back-filled from the record when the file was created.

The last column of Table 12 is taken from the record files named in each row's source comment, because the log does not always say who caught an error. Dates are local dates.

Table 12. Claude's weaknesses W1 to W13, with dated instances (at most three per row), the countermeasure adopted and who caught the failure.

#	Weakness	Dated instances	Countermeasure	Who caught it
W1	Loses the big picture to local polish	c17, 8 Oct: Claude's own warning that caution had become the product. c27 to c35, 12 to 18 Oct: led the work into one-dimensional curve worlds. 19 Oct: offered fallback options that each measured one piece of research	Erin's North Star kept as their verbatim words in the standing project file; scope protocol; status file read every session	Erin (18 Oct and 19 Oct)
W2	Re-proposes old ideas as new	c23, 11 Oct: "Discovery Horizon" was the c17 "Jump Horizon". 18 Oct: nearly framed "unification worlds" as new. c46, 21 Oct: option A left out the c10 to c15 record	Search the record before any proposal; cite the earlier round	Claude itself (c25 table; 18 Oct); its own review pass (c46)
W3	Conclusions outrun the evidence	c21 and c30, 10 and 17 Oct (back-filled): "first measurement", "G is bounded", "contamination-proof". 19 Oct: "Sol solved 48 of 49" stated without the note that at least 20 were AXIOM, HINT or control cells. c44, 21 Oct: "non-self-averaging observables are never bias-limited"	Prove a property before naming it; copy headline claims from the primary report with its qualifiers	Pro (c22 for c21, withdrawn by Claude in c23; c31 for c30; c45 for c44); Claude's own failure table (c25) also listed the c21 items; a package review (19 Oct)
W4	Vacuous or ambiguous rules in specs	c32, 18 Oct: the "blind" floor author saw the laws-given cards (A2). c32 and c33: the CLUE UNUSED anchor was vacuous (P3). c34, 18 Oct: N3 named log-quantiles, not probabilities	Test each rule on synthetic outcome patterns before freezing; name every output a tolerance covers	A2: Claude, in a note to Pro. P3: Claude, after the pilot. N3: Codex, in its Windows replay
W5	Acts on an unchecked assumption about what Erin means	18 Oct: took Erin's request for a new chat to mean a Claude chat; Erin had meant ChatGPT Pro. 18 Oct: proposed a seven-model ladder that included the expensive Astra model	Ask in one line when the app, model or budget matters	Erin (the chat); Erin's model rule (the ladder)
W6	Does not predict saturation; plans runs that change no decision	18 Oct: planned a 16-episode Luna screen of c35 links it had predicted were too easy. 19 Oct (c37 brief): asked Pro for research before any feasibility arithmetic	Write the prediction first; name the decision a run could change; run the cheap admission gates before research	Erin (18 Oct); Claude's c38 check (19 Oct)

#	Weakness	Dated instances	Countermeasure	Who caught it
W7	Famous templates can pass for discovery	c17, 8 Oct: the v0.1 worlds were Planck, special relativity and general relativity in disguise. c35, 18 Oct: T3 examples were textbook identities	Generate structure from held-out grammars; no famous templates	Claude, reading Sol's result on v0.1 (c19, 9 Oct); Claude withdrew its "recognition" reading in c21, because a trace audit could not tell recall from derivation. No separate catch logged for T3
W8	Verbosity and density bury the key point	Undated in the log: Erin said Claude's replies were too dense and technical	Plain summary first; tables; depth in files	Erin (the log records no further catch)
W9	Agrees too fast with a strong reviewer or the user	c46, 21 Oct: a first draft recommended Pro's option A over Claude's own c17 record, with no new evidence	Restate the evidence; keep contrary evidence	Claude's three-reviewer pass
W10	Forgets across context limits	19 Oct: Erin restated the goal because it had dropped out of Claude's context at compaction. No other dated instance is logged	Keep state in files, not in context	Erin (19 Oct)
W11	Procedural inertia: repeats a step because it was done last turn	18 Oct: kept "verify, then Luna screen, then Sol" for the c35 material after the direction had changed. 19 Oct: added the new rule to its own files but not to Pro's brief	Ask whether each step is needed now; search every outgoing message when a rule changes	Erin (18 Oct); a package review (Pro's brief)
W12	Gates the real test behind a proof that cannot be given	c38 and c39, 19 Oct: the critic brief and the c39 message asked that programs fail while a legal witness succeeds. c39, 19 Oct: Pro required an attainable improvement before another build; c40 interpreted this as reinstating the witness problem	Programs reject, models certify; the smallest model pilot follows the kill gate	Pro (c39); Claude, Fable 5.1 turn, named it (c40, 20 Oct)
W13	Treats a compute ratio as research headroom	c41 and c42, 20 Oct: "scale until the standard method falls short"; the N24 gap was hardware speed. c43 and c44, 20 and 21 Oct: a reach ratio, then G of at least 20 against an unengineered floor. c47 and c48, 21 Oct: a variance-ratio score that could not certify bias	Measure G at the scored precision against an engineered floor; adversarial review of every new threshold	Pro (c41, c43, c47); Claude's review pass (c44); Claude's probe (c48)

The strategic memo (ROAD_AFTER_C39) asked why the rounds from c17 to c39 had produced no model episode on any discovery design (the memo counts 21 rounds but gives no list). It separates these rounds from the early single worlds, the landing tests, the clue pilot and the c10-c15 campaign, which had model data. In our terms, the coupled or later discovery candidates (the c24 coupled lab, c37, the c38 repairs D1 and D2, and c39 F1 and F3) had no model run, while v0.5, v0.6, T1 and M5 did run (Appendix B). The memo's reading, offered as "not a theorem", was that the admission rule demanded that a cheap generic program fail while a legal witness (a solver the task's rules allow, used to show the task is solvable) succeed. For computable truth, a succeeding witness is itself a program. So, if "generic" covers any program a competent solver could write, the rule asks that a program both exist and not exist (Section 6.5 gives the narrower reading).

The weakness log repeats the memo's count of 21 rounds in the span c17 to c39 (with no list; continuation numbering has gaps and sub-rounds) and notes Sol's 7 committed successes in 8 in the c10 to c15 campaign. Stated precisely, only the coupled or later discovery candidates and everything proposed from c37 (19 Oct) on had no subject-model run (Appendix B).

In Table 12, the failures about the goal and the plan (W1, W5, W6, W8, W10, W11) were mainly caught by Erin; W6 and W11 also list Claude's own c38 check and a package review. Most failures about claims, rules and arithmetic (W3, W4, W9, W12, W13) were caught by Pro, Codex or Claude's own review passes. This is an observation about the rows as logged, not a rate.

Table 13 lists twelve failures of Pro and Codex. It is a selection, not a count, chosen to span the project and the kinds of error.

Table 13. Notable failures of Pro and Codex, with the round and who caught them.

What	Round	Caught by
Codex: during the headroom study, coordinator metadata that held one attempt's public score was written into another attempt's scratch folder	Headroom study, 25 Sep	Codex itself, which recorded it. Non-exposure rests on the solver's self-report
Pro: named a quantum-placement improvement target without running the published reference, which already scored 186/186	c07, 26 Sep	Pro, when it ran the reference the next turn
Pro: called its cost-75 design the strongest generic result; a cost-focused ordinary optimizer reached 67	c08, 2 Oct	Codex
Codex: after Docker was switched off, the operator stopped at a blocker; Erin judged that the interrupted slot should have resumed, while the frozen protocol forbade automatic retries. Whether resuming fell under that rule is unresolved (a disputed case)	Campaign 14, 4 Oct	Erin
Pro: labelled a 128 times stress cutoff as the inherited v0.1 cutoff (the actual value was 10,000 times) and issued a 6.8% defect warning	c20, 9 Oct	Pro's own QA, which withdrew the warning
Pro: the c31 release wrapper failed on an import error before the ZIP was made. 81 tests had passed, but no code survived, only 10 documents	c31, 17 to 18 Oct	Pro disclosed it; Claude established what survived
Pro: the c37 admission code checked only whether its own table's loss was zero; its own switching rule had fired while the next-round note still proposed stronger dependence	c37, 19 Oct	Claude (c38); Pro conceded in c39
Pro: required a useful, attainable improvement over a current reference before a new benchmark build; Claude judged in c40 that this reinstated the witness problem (the c39 text did not itself specify an in-budget scripted witness)	c39, 19 Oct	Claude (c40, 20 Oct)
Pro: the c41 Windows meter charged 0.015625 CPU s for a worker that used 60.6 s, because it timed only the launcher process	c41, 20 Oct	Claude (c42). No decision changed, because the worker stopped itself
Codex: its reach numbers were too small. Its largest passing 2D Ising size was 512 (fitted 1,024); Claude's tuned code reached 2,048 in 171 CPU s. Codex's code was correct	Codex set 2, 20 to 21 Oct	Claude (c44)
Pro: the registered c45 ladder assumed the memory depends on g^2t, which does not hold at g = 0.01 to 0.08	c45, 21 Oct	Claude's independent test (c46)
Pro: fixed the pilot evaluation sizes L = 8 and 10 with no saturation finding (its plan allowed a move to L = 12 and 16 after one). A plain parallel-tempering code (no cluster moves) reached the same lowest energy in all four runs on every L = 8 and L = 10 packet, the best found within 3.44 s of a 30 s budget	c47, 21 Oct	Claude (c48)

E.1 The recurring modes in detail

These are the six recurring modes summarized in Section 7, with one or two sentences each; the episodes are told in the phases of Section 4.

1. **Conclusions outrunning evidence (W3).** On 10 October (c21), Claude certified that two rival laws agreed within 0.08% inside the laboratory. The check used temperatures outside the legal range [1.232, 15.3], and the in-range gap was 0.352%. Pro found this in c22, and Claude withdrew the claim in c23, together with a "first measurement" claim pre-empted by the close prior work ULTRADISCOVERY. The tally "48 of 49" appeared without its qualifiers in c25, and again in a package

for Pro, until a 26-agent review on 19 October (20 findings kept, 3 refuted) restored them. Pro had announced a 6.8% inherited-cutoff concern, computed against a 128-times cutoff when the inherited cutoff was 10000 times, and withdrew it in c20.

2. **Losing the goal, and re-proposing old ideas (W1, W2).** After c17 diagnosed that "caution became the product", Claude led c27 to c35 into narrow curve worlds with heavy gating. In c23 it presented the "Discovery Horizon" as new, although it was its own c17 "Jump Horizon". On 21 October (c46), its first draft of option A omitted the c10 to c15 campaign, in which Sol at max committed 7 of 8 successes, until a review caught the omission. Pro wrote of itself: "We improved the machinery faster than we improved the research question."
3. **Designer leaks and vacuous rules (W4).** The v0.1 noise level revealed the hidden speed C, which Claude's leak test, a check of literal equality only, had missed. Pro's audit found it, and Claude accepted it in c19. The v0.3 "connection" questions (U2) could be answered from one hall, and Codex's gate excluded all five U2 cells before any model run. A frozen T1 rule (CLUE UNUSED) had an always-true clause because the "blind" floor author had seen the L1 cards.
4. **Procedural inertia (W6, W11).** On 18 October, Claude kept a "verify, then Luna screen, then Sol" sequence for c35 after the direction had changed. It planned a 16-episode Luna screen that it expected to be too easy, and whose result could change no decision. Erin cancelled the screen (c35 itself was superseded earlier, when the old Pro chat stalled), saying a step should depend on the circumstances of the moment, not on having been done last turn. The lesson became W11.
5. **Demanding certainty before acting (W12).** From c17 to c39, no subject-model episode ran on any of the coupled or later discovery candidates (the c24 coupled lab, c37, the c38 repairs D1 and D2, and c39 F1 and F3), and none ran on any design proposed from c37 on. Model data from those rounds exist only for the early single worlds, the landing tests (v0.5 and v0.6) and the T1 clue pilot. In our reading, one cause was an admission rule that required a cheap program to fail while a legal witness succeeded (Section 6.5). An 18 October rule already said that a design round must be followed by model data, and c37 to c39 still produced none. Pro's c39 recommendation required a useful, attainable improvement over a current reference before another benchmark build. Claude interpreted this in c40 as reinstating the witness problem, although c39 did not itself specify an in-budget scripted witness, and a Fable 5.1 turn named the trap on 20 October.
6. **Treating a compute ratio as research headroom (W13).** This happened three times. On 20 October (c41), the closest quantum case missed its 60 s cap only on speed: 61.1 CPU s uncapped here (the capped worker stopped itself at 60.6 s) and 54.7 s on Pro's host. A direction to scale until the standard method falls short had invited this, and Pro objected first. On 21 October (c43/c44), Claude specified the gap as a reach ratio, ignoring that the key must be $(\tau/r)^2$ times more precise and that an engineered floor can be several times faster. Its first c44 draft proposed $G \geq 20$ against an unengineered floor. Pro had registered the precision-aware definition first, and three reviewers caught the draft. On 21 October (c47/c48), Claude proposed a variance-ratio efficiency score, and Pro's exact counterexample showed a variance gain of 100 with a worse mean-square error. A probe then found that one plain parallel-tempering code reached the same lowest energy in all four runs on every $L = 8$ and $L = 10$ packet, within 3.44 s of a 30 s budget (Section 4.5, Section 4.6).

Infrastructure was a seventh source of lost rounds, though not a reasoning failure. Pro's c31 sandbox expired, leaving 10 design documents, and the old Pro chat stalled for about 4 hours during the c35 build.

E.2 Cross-checks and reviews

Cross-checking between agents. Pro audited Claude and found the noise leak (c18), the 0.08% error (c22) and the contradiction in the witness gate (c39), and killed the variance score (c47). Claude verified Pro natively and found a spec-only learner that scored 576/576 (c38), a Windows meter that charged

0.015625 CPU s for a 60.6 s worker (c42) and a rescaling assumption that did not hold (c46). Codex's gates stopped the U1 and U2 leaks before any model saw them.

Adversarial reviews caught drafts before they went to Pro: 26 agents on 19 October, and three reviewers each in c44 and c46 (the c46 review also covered the message for c47).

Appendix F AI use, authorship and data

Authorship. Erin Krasniqi set the goal and the resource constraints, authorized every model run, relayed work between the AI systems, retained final authority over major changes of direction, and caught several strategic errors of the AI systems (Section 7). Erin is responsible for this paper.

Erin's contribution was also intellectual, not only managerial. Their opening brief of 23 September already asked whether an AI could build a benchmark that would be hard even for itself, the question Section 6.6 returns to. Their earlier notes on physics intuitions, in which they proposed hypotheses from little information, were the team's working example of the "jump" it wanted to measure. Reading them on 25 September, Pro singled out three operations: changing how a problem is represented, proposing that separate categories are one, and asking what the current description no longer shows. These shaped the designs: Claude's c17 definition of the jump (Section 4.2), the hidden-unity worlds of v0.3 (Pro's c22 review), and requirement (f) of c41, noticing incompleteness (Section 4.5). Erin also proposed treating the AI team's own failures as evidence, which became Section 7, and supplied the criterion for which of those failures could become benchmark targets: overcoming the weakness must be part of doing science.

AI systems and their roles.

- **ChatGPT Pro** (GPT-6 Pro): designs, mathematics, literature research and prototypes; Pro led Phase 1 and, from c37 on, most design rounds.
- **Codex** (the codex-cli agent on Erin's computer): built the harnesses, ran all subject-model episodes, and measured hardware reach.
- **Claude** (Opus 5.5, with one turn by Fable 5.1): set direction from 8 October, wrote specifications and messages, and verified results independently.
- **Tested models:** gpt-6.1-sol and gpt-6-luna, plus gpt-6-sol and GPT-6 Astra in September.

How this paper was written. The paper was written after the work ended, partly by the AI system that did much of it, so we built it from the record, not from memory. Stage 1 was a workflow of 33 agents: 16 chronicle readers (one slice of the project each), 16 independent fact-checkers who re-opened the primary source for each claim (973 claims: 942 confirmed, 26 corrected, 5 unsupported), and one completeness critic, who listed 14 gaps. In stage 2 Claude (Opus 5.5) wrote the thesis, outline, abstract, introduction, analysis, proposals, limitations and conclusion; Claude Sonnet 5.5 subagents drafted most other sections, each re-opening the primary file for every number, and adversarial review and revision followed. Pro then reviewed the full draft adversarially twice (c49 and c50), and the corrections were applied. A writer that is also the subject invites overclaiming, so these steps matter.

Erin chose the write-up on the night of 21 October, approved this version for release, and remains the sole human author. Earlier unrelated AI chats of Erin's are not described here.

In the project's source files, every number carries a comment naming the record file it was copied from. Erin approved the use of their full name; their messages are paraphrased rather than quoted. The analysis in Section 6 and the proposal in Section 8 were written by Claude and should be read as the team's interpretation.

Data. The project record is private at the time of writing. It holds the round folders, verification reports, model-run outputs and chat exports. It contains material unrelated to this work, which would be removed before any release. A redacted release of the round folders, the verification reports and the probe code (for example, the spin-glass saturation probe of Appendix D) is possible at the author's discretion.

Cost. No paid APIs were used; runs used existing subscriptions.